

UNIVERSITÀ DI PISA



CORSO DI LAUREA IN MATEMATICA

**From Theory to Real World:
Simulating the Spread of Infectious
Diseases with the SIRCD Model**

TESI DI LAUREA MAGISTRALE
IN MATEMATICA

CANDIDATO

Andrea Tedesco

RELATORI

Prof. Vladimir Georgiev

Dott.ssa Maria Laura Manca

CONTRORELATORE

Prof. Jacopo Bellazzini

Università di Pisa

ANNO ACCADEMICO 2023 - 2024

Alla mia famiglia

Contents

Contents	3
Acknowledgements	5
Introduction	9
1 Presentation of a SICRD model	13
1.1 Mathematical modelling	13
1.2 Qualitative analysis of the model	15
1.2.1 Positiveness of the solution	15
1.2.2 Equilibrium points	17
1.3 Introduction of vaccination	20
1.3.1 Positiveness of the solution	21
1.3.2 Qualitative analysis of the model	22
1.4 Possible improvements of the model	24
1.4.1 Time-Dependent Parameters Extension	24
1.4.2 Age-Based Extensions	26
2 Introduction of diffusion	29
2.1 Regional kinetics model	29
2.1.1 Qualitative analysis of the multi-regional model	30
2.2 Global existence of the solution	33
3 Numerical simulations	39
3.1 Simplification of the model	39
3.2 Presentation of the algorithm	40
3.3 Results	51
3.4 Conclusions	55
4 Appendix A	59
4.1 Theory of Ordinary Differential Equations	59
5 Appendix B	73

5.1	Time-dependent reproduction number	73
5.1.1	Interpratation of reproduction number	74
Bibliography		77

Acknowledgements

A heartfelt thanks to Professor Vladimir Georgiev and Doctor Maria Laura Manca, supervisors of my thesis, for their constant support, the essential tips, and the complicity in the realization of each chapter of my work.

A heartfelt thanks to the academic community of the Department of Mathematics of the University of Pisa, for providing me with a stimulating environment rich in opportunities for personal and professional growth.

Thanks to my parents who supported me on every occasion, especially in the darkest moments, who supported me and comforted me in the periods of greatest pressure, who gave me the strength to move forward, and who always encouraged me to continue on the path I took, to cultivate my passion with resilience, to look at the finish line without considering the time.

To my brother Marco who was an example to me for dedication, sacrifice, and willpower and helped me settle into a new reality different from the one I was used to by taking care of me.

To my furry Aura which in these years has always welcomed me wagging her tail when I returned home. Sweet face and eyes, princess of my family, a natural anti-stress.

To my grandmother who, at every moment, never failed to give me attention, affection, and generosity despite the distance that separates us.

To my cousin Francesco who was close to me when things weren't going well, who was able to motivate me, despite my mental and emotional energy being lacking, who was always ready to make me laugh with his humor.

To my uncles who encouraged me and who were able to give me tips on my path.

To my grandparents who are no longer here, but who are beside me smiling. I thank all the friends I met here at university, with whom I shared this environment, this city, and a thousand other experiences, which allowed me to unplug and face obstacles lightly. I couldn't have wished for a better group of people to live this adventure with. Some have shared with me only the beginning of this journey, some only the final part and some have been present from the beginning to the end, but I am grateful to each of them for the unforgettable evenings, lunches, dinners, trips out of town, days of study and relaxation.

I thank my long-time friends who I have known for practically 20 years. They are lifelong companions more than friends. I have spent the most intense moments and shared joys and sorrows with them. I am infinitely grateful to them for all the times they have been understanding towards me when I declined their proposals to go out for a few more hours of study. I will be eternally grateful to all of them for the lightness with which they face life, which helps me remember how important it is to continue to cultivate certain bonds, which have now become indissoluble. Even if when we are reunited we seem ruined because of the carefree way we live our days, we remain loyal and present towards each other since we know how important the value of friendship is. Thank you for all the serious meals, the green ones at late hours, the rides, the moments of leisure and stupidity shared that have a different flavor with you.

I thank my high school friends for the laughter and good times spent in recent years. I also owe them an apology for the numerous refusals to go out for a few more hours of study and sleep. I thank them for their understanding and for their immense kindness towards me even though as the years go by, the times we see each other are fewer and fewer. Thank you for the magnificent times spent together and for making some moments of my life memorable. Thank you for the words of comfort, the advice, and the tips that I have treasured and that I will always carry with me.

To seal these thanks, my special thought goes to Dania, the person who more than anyone else has certainly been able to inspire me, support me, and put up with me. These few lines are not enough to express all my gratitude towards her. She has transmitted to me the spirit of sacrifice and a great sense of responsibility. For me, she has represented an immense source of inspiration that has allowed me to mature and move forward in the most complicated moments. She has helped me grow both as a student and as a man and has accompanied me for much of my journey. I am infinitely grateful to her because thanks to her I have discovered my childish side, because with her I am able to bring out the best part of me, because, even though I have known her for only 6 years, we have built an inseparable relationship in which I have learned so much, I still have so much to learn and certainly I will always have more.

And then, to Dania's parents, my dear and grateful thought. For them, I have a deep feeling of gratitude. They welcomed me and loved me from the first moment as if I were a son.

Thank you all! Today is the last stage of my university journey and now I look forward.

Un sentito grazie al Professor Vladimir Georgiev e alla Dottoressa Maria Laura Manca, relatori della mia tesi, per il supporto costante, le dritte indispensabili e la complicità nella realizzazione di ogni capitolo del mio lavoro.

Un grazie sentito alla comunità accademica del Dipartimento di Matematica dell'Università di Pisa, per avermi fornito un ambiente stimolante e ricco di opportunità di crescita personale e professionale.

Grazie ai miei genitori che mi hanno sostenuto in ogni occasione, specialmente nei momenti più bui, che mi hanno supportato e confortato nei periodi di maggiore pressione, che mi hanno dato le forze per andare avanti, che sempre mi hanno incoraggiato a continuare la strada intrapresa, a coltivare la passione con resilienza, a guardare il traguardo senza considerare il tempo.

A mio fratello Marco che mi è stato d'esempio per dedizione, sacrificio e forza di volontà e che mi ha aiutato a insediarmi in una nuova realtà diversa da quella a cui ero abituato prendendosi cura di me.

Alla mia pelosetta Aura che in questi anni mi ha sempre accolto scodinzolando al mio ritorno a casa. Musetto e occhi dolci, principessa della mia famiglia, un antistress naturale.

A mia nonna che, in ogni momento, non mi ha fatto mancare mai attenzione, affetto e generosità nonostante la distanza che ci separa.

A mio cugino Francesco che mi è stato vicino quando le cose non andavano per il meglio, che mi ha saputo motivare, nonostante le energie mentali ed emotive venissero a mancare, che è sempre stato pronto a farmi ridere con il suo umorismo.

A miei zii che mi hanno spronato e che hanno saputo darmi dritte sul mio percorso.

Ai nonni che non ci sono più, ma che mi sono accanto sorridenti.

Ringrazio tutti gli amici che ho conosciuto qui all'università, con i quali ho condiviso questo ambiente, questa città e altre mille esperienze, che mi hanno permesso di staccare la spina e di affrontare gli ostacoli con leggerezza. Non avrei potuto desiderare un gruppo di persone migliore con cui vivere questa avventura. Alcuni hanno condiviso con me solo l'inizio di questo percorso, alcuni solo la parte finale e alcuni sono stati presenti dall'inizio alla fine, ma sono grato a ognuno di loro per le indimenticabili serate, i pranzi, le cene, le gite fuori porta, le giornate di studio e di relax.

Ringrazio i miei amici storici che conosco da praticamente 20 anni. Compagni di una vita più che amici. Con loro ho trascorso i momenti più intensi, con loro ho condiviso gioie e dolori. Sono infinitamente grato a loro per tutte le volte che sono stati comprensivi nei miei confronti quando declinavo le loro proposte di uscita per qualche ora di studio in più. Sarò eternamente riconoscente a tutti loro per la leggerezza con cui affrontano la vita che mi aiuta a ricordare quanto sia importante continuare a coltivare certi legami, diventati ormai indissolubili. Anche se quando siamo riuniti sembriamo dei rovinati per via della spensieratezza con cui viviamo le giornate, rimaniamo leali e presenti

l'uno nei confronti dell'altro dal momento che sappiamo quanto sia essenziale il valore dell'amicizia. Grazie per tutte le mangiate serie, le verdi a ora tarda, i passaggi, i momenti di svago e di stupidità condivisi che con voi hanno un altro sapore.

Ringrazio gli amici del liceo per le risate e i bei momenti passati negli ultimi anni. Anche a loro devo delle scuse per i numerosi rifiuti alle uscite per qualche ora di studio e di sonno in più. Li ringrazio per la comprensione e per l'immensa benevolenza nei miei confronti nonostante col passare degli anni siano sempre meno le volte in cui ci vediamo. Grazie per i magnifici trascorsi assieme e per aver reso memorabili alcuni momenti della mia vita. Grazie per le parole di conforto, i consigli e le dritte di cui ho fatto tesoro e che porterò sempre con me.

A suggello di questi ringraziamenti il mio pensiero speciale è per Dania, la persona che sicuramente più di ogni altra è stata in grado di ispirarmi, sostenermi e sopportarmi. Non bastano queste poche righe per esprimere tutta la mia gratitudine nei suoi confronti. Mi ha trasmesso lo spirito di sacrificio e un grande senso di responsabilità. Per me ha rappresentato un'immensa fonte di ispirazione che mi ha permesso di maturare e di andare avanti nei momenti più complicati. Mi ha fatto crescere sia come studente sia come uomo e mi ha accompagnato per gran parte del mio percorso. Le sono infinitamente grato perché grazie a lei ho scoperto il mio lato fanciullesco, perché con lei riesco a tirare fuori la parte migliore di me, perché, nonostante io la conosca da soli 6 anni, abbiamo costruito un rapporto inscindibile in cui ho imparato tanto, ho ancora tanto da imparare e di certo ne avrò sempre.

E, poi, ai genitori di Dania il mio pensiero caro e di gratitudine. Per loro nutro un profondo sentimento di riconoscenza. Mi hanno accolto e voluto bene sin dal primo istante come se fossi un figlio.

Grazie a tutti! Oggi è l'ultima tappa del mio percorso universitario e adesso guardo avanti nella speranza di vivere altri momenti significativi confidando di poter contare ancora su tutti voi.

Introduction

The spread of infectious diseases has always been a critical area of study in epidemiology, especially in the wake of global pandemics. The SIR model is a fundamental framework in epidemiology used to understand the spread of infectious diseases within a population. Developed by Kermack and McKendrick in 1927, the model divides the population into three compartments:

- Susceptible (S): Individuals who have not yet contracted the disease but are at risk of becoming infected.
- Infected (I): Individuals who have contracted the disease and are capable of spreading it to susceptible individuals.
- Recovered (R): Individuals who have recovered from the disease and are assumed to have gained immunity, thus no longer susceptible to infection.

This dissertation delves into the intricacies of the SIRCD model, designed to simulate the dynamics of infectious diseases over time and across geographical regions. The SIRCD model extends the classic SIR model by incorporating compartments such as carriers and deceased individuals, providing a more comprehensive understanding of disease progression.

- The carriers compartment represents individuals who are infected with the disease but do not show symptoms. These asymptomatic carriers can still transmit the disease to susceptible individuals. Including a carrier compartment helps to model the hidden spread of the disease, which is particularly important for diseases where asymptomatic transmission is significant.
- The deaths compartment accounts for individuals who have succumbed to the disease. This compartment is crucial for understanding the mortality impact of the disease and for public health planning. It helps to differentiate between those who recover and those who do not survive the infection.

In the first Chapter, we introduce the SIRCD model, meticulously describing each compartment—Susceptible, Infected, Recovered, Carrier, and Deceased—and the interactions between them. We explore the fundamental quantities and parameters that govern the model, setting the stage for a qualitative analysis of the system of differential equations that underpin it. This analysis includes examining equilibrium points, stability, and the behavior of solutions, ensuring a robust theoretical foundation.

Recognizing the need for realism in epidemiological modeling, we propose several modifications to the basic SIRCD framework. These enhancements include the introduction of a vaccination campaign, which simulates the impact of immunization on disease spread, and dividing the population into age groups to account for varying susceptibility and recovery rates. Additionally, we extend the model to incorporate time-dependent parameters, reflecting the dynamic nature of real-world epidemics.

A significant portion of the second Chapter is dedicated to the geographical spread of infectious diseases. We introduce a transport factor that models the movement of individuals between different regions, thereby simulating the spatial dynamics of an epidemic. This addition is crucial for understanding how diseases propagate across populations and borders. We also present the theorem of global existence and uniqueness of the solution, proving it under the assumption of minimal initial data. This theoretical result is pivotal for ensuring the reliability and accuracy of our model.

In the third Chapter, we shift our focus to practical applications, specifically the COVID-19 epidemic in Italy. We develop an algorithm to derive the parameters necessary for our model to accurately represent the epidemic's progression. By comparing the model's predictions with real-world data, we validate its effectiveness and highlight areas for improvement. Graphical representations of the results provide a clear and intuitive understanding of the epidemic's dynamics, showcasing the model's ability to capture the nuances of disease spread.

Appendix A offers a comprehensive overview of the mathematical tools and concepts essential for understanding the dissertation. This includes a detailed discussion of Ordinary Differential Equations (ODEs), stability analysis according to Lyapunov, the fixed point theorem, and relevant aspects of linear algebra. These foundational elements are crucial for appreciating the depth and rigor of the analyses presented in the main chapters.

Finally, in Appendix B there is a brief summary of the meaning of the time-dependent reproduction number. Through the code, we can also graphically represent its trend, which allows us to better understand the real course of the epidemic, namely the decline, and expansion.

In conclusion, this dissertation provides a solid theoretical and practical foundation for modeling the spread of infectious diseases, offering both qualitative analyses and real-world applications. Although some simplifications and lim-

itations—such as the availability of accurate data—prevent exact predictions, the model demonstrates a strong ability to approximate the epidemic’s progression and capture its key dynamics.

In addition to the theoretical and practical insights provided by this dissertation, an original contribution of this work lies in the development of a novel algorithm for parameter estimation. Unlike the method presented in the main article, the algorithm implemented here was designed specifically to optimize the fit of the SIRCD model to real-world data, addressing some of the limitations inherent in existing approaches. This custom algorithm allows for greater flexibility and accuracy in capturing the complex dynamics of the epidemic, representing a significant advancement in the modeling process. By combining this computational innovation with theoretical analyses and practical applications, this dissertation offers a comprehensive framework for understanding and predicting the spread of infectious diseases.

The work presented here opens up several directions for future research. The SIRCD model can be further refined by including additional compartments, such as hospitalized individuals or reinfected cases, to better reflect real-world dynamics. The integration of stochastic elements could help analyze randomness in disease transmission while coupling the model with real-time spatial data would enhance its geographical accuracy. Furthermore, applying machine learning methods could improve parameter estimation and predictive capabilities. Expanding the model to account for factors like immunity waning, vaccination campaigns, or behavioral changes over time could provide a more complete understanding of epidemic evolution. These developments hold significant potential to support public health decision-making and improve preparedness for future infectious disease outbreaks.

Presentation of a SICRD model

In this chapter, we will present and analyze a mathematical model that simulates the spread of an infectious disease. In general, the SIR (susceptible-infectious-recovered) model is a well-known model for predicting the number of people (or animals) in a population who become infected by and then recover from a disease. It is possible to make modifications that include categories such as people who have been exposed to the disease but are not yet infectious or those who die from the disease.

1.1 Mathematical modelling

At first, let us consider a single region where an individual within the population can be classified as susceptible (never had the disease), infected with the disease, recovered from the disease (and who is assumed to be immune), or have died. We take inspiration from a model in the Article [5, Section 2.1]. Let S , I , R , and D denote the total number of individuals in the population who are susceptible, infected, recovered, or have died respectively

$$\begin{aligned}
 \frac{dS}{dt} &= -\lambda SI \\
 \frac{dI}{dt} &= \lambda SI - \mu_{IR}I - \mu_{ID}I \\
 \frac{dR}{dt} &= \mu_{IR}I \\
 \frac{dD}{dt} &= \mu_{ID}I
 \end{aligned} \tag{1.1}$$

where λ is the infection rate, that we can assume to be $0 < \lambda \ll 1$ since at the start of an outbreak, the total number of infected individuals is typically small. As a result, there are fewer interactions between susceptible and infected individuals, reducing the overall number of new infections. In some models, simplifying assumptions (like supposing λ is small) helps create analytical solutions or allows for easier qualitative analysis. While the assumption may not reflect reality precisely, it can help study specific properties of the system. μ_{IR} is the rate at which infectious people recover and μ_{ID} is the rate at which infectious people die. We note that the sum of the right-hand sides of the differential equations is zero, indicating that there will be no change in the size of the population.

However, this model does not apply to diseases where people may be infected with the disease but have no symptoms and are unaware that they are infected. Here, we will refer to these people as carriers. Hence, the spread of a disease may be more rapid and widespread in a population than is indicated by the number of infected people. This can be incorporated into the basic SIRD model with including an additional class, C , of carriers who are either infected with the disease but do not have any symptoms or who have very mild symptoms and so do not get tested for the disease. It is important to note and emphasize here that both infected individuals and carriers can infect other people as both are infected with the disease and the only difference is that carriers are unaware that they are infected. Including the carrier category leads to the modified system of differential equations, for which we drew inspiration from [5, Section 2.1]. Unlike the previous model, we omit the equation for the deceased people. We will introduce the class of dead people subsequently. The model is the following:

$$\begin{aligned}\frac{dS}{dt} &= -\lambda S(I + C) \\ \frac{dI}{dt} &= \lambda \beta S(I + C) - \mu_{IR}I - \mu_{ID}I \\ \frac{dC}{dt} &= \lambda(1 - \beta)S(I + C) - \mu_{CR}C - \mu_{CD}C \\ \frac{dR}{dt} &= \mu_{IR}I + \mu_{CR}C\end{aligned}\tag{1.2}$$

where μ_{CR} is the rate at which the carriers recover from the disease and cease to be infectious, μ_{CD} is the death rate for carriers and β is the proportion of the population who become infected with the disease, have symptoms and are diagnosed as having the disease. It follows $1 - \beta$ is the proportion of the population who become infected with the disease but who are not diagnosed as having the disease, either because they do not have symptoms, they have mild symptoms and mistake the disease under consideration for another disease, or they simply refuse to get a diagnosis.

1.2 Qualitative analysis of the model

1.2.1 Positiveness of the solution

Before studying the equilibrium points, we can note that the solutions of the system (1.2) are non-negative for every t in the existence interval, if we assume that the initial conditions are positive.

Let us start from the first equation of (1.2). By assuming that $S(0) > 0$, we want to say that $S(t)$ will be non-negative for every t in the existence interval. The differential equation for $S(t)$ is given by:

$$\frac{dS}{dt} = -\lambda S(I + C),$$

where $\lambda > 0$ is a constant and $S(0) > 0$.

Assume that $S(t)$ becomes non-positive at some finite time $t_1 > 0$. That is:

$$S(t) > 0 \quad \text{for } t \in [0, t_1), \quad \text{and } S(t_1) = 0.$$

From the differential equation, at t_1 , substitute $S(t_1) = 0$:

$$\left. \frac{dS}{dt} \right|_{t=t_1} = -\lambda S(t_1)(I(t_1) + C(t_1)).$$

Since $S(t_1) = 0$, this simplifies to:

$$\left. \frac{dS}{dt} \right|_{t=t_1} = 0.$$

Thus, $S(t)$ reaches zero at t_1 with a zero derivative, meaning it cannot decrease further. For $t \in [0, t_1)$, $S(t) > 0$. Thus, the differential equation becomes:

$$\frac{dS}{dt} = -\lambda S(t)(I(t) + C(t)).$$

If $I(t) + C(t) \geq 0$, then $\frac{dS}{dt} \leq 0$, meaning $S(t)$ is non-increasing. Importantly, as long as $S(t) > 0$, $\frac{dS}{dt}$ cannot cause $S(t)$ to become negative, because the right-hand side of the equation depends linearly on $S(t)$, which is zero at t_1 . At t_1 , $S(t_1) = 0$, and $\frac{dS}{dt} = 0$. However, for $S(t)$ to become negative for $t > t_1$, $\frac{dS}{dt}$ would need to be negative at t_1 . This is not possible because:

$$\frac{dS}{dt} = -\lambda S(t)(I(t) + C(t)).$$

When $S(t_1) = 0$, the entire right-hand side evaluates to zero. Therefore, $S(t)$ cannot decrease further into negative values. Thus, $S(t) \geq 0$ for all $t \geq 0$.

If we prove that also $I(t)$ and $C(t)$ fulfill this property, then it will be the same for R . If $I, C > 0$ then R will be positive. Hence, since the positive derivative

implies that the function grows, R will remain positive.

So the last claim is the positiveness of I and C . The proof is pretty similar to the previous one. If by contradiction $I(t) < 0$ for any t , then, due to the Sign Permanence Theorem and the continuity of I , there exists t_0 such that $I(t_0) = 0$ and, after t_0 , I is negative. Now we define

$$\Omega_I = \{t \in (0, t_0] \mid I(t) = 0, \exists \delta > 0 \text{ such that } I(s) < 0 \forall s \in (t, t + \delta)\}$$

and $t_m := \min \Omega_I$, that is well defined as Ω_I is not empty ($t_0 \in \Omega_I$), I is a continuous function and $I(0) > 0$. We note that

$$\dot{I}(t_m) = \lambda \beta S(t_m) C(t_m),$$

so the sign of the right-hand side is determined by that one of C , because of the positiveness of λ, β and S , previously showed. If $C(t_m) \geq 0$, then $\dot{I}(t_m) \geq 0$, therefore I will not decrease and this fact is in contradiction with $I(s) < 0$ for every $s \in (t_m, t_m + \delta]$. Instead, if $C(t_m) < 0$, then $\dot{I}(t_m) < 0$. At this point, we consider the equation of C . We define

$$\Omega_C = \{t \in (0, t_m] \mid C(t) = 0, \exists \delta > 0 \text{ such that } C(s) < 0 \forall s \in (t, t + \delta)\}$$

and $t_k := \min \Omega_C$. Since $t_k \leq t_m$, $I(s) \geq 0$ for every $s \in (0, t_k]$. So we get

$$\dot{C}(t_k) = \lambda(1 - \beta)S(t_k)I(t_k) \geq 0,$$

that is in contradiction with the definition of t_k , as the non-negative derivative implies the non-degrowth of the function. In conclusion, I will remain positive, if the initial condition $I(0) > 0$. By applying the same argument, we can prove the positiveness of C .

Now, if we define $Y(t) = S(t) + I(t) + C(t) + R(t)$, then we get

$$Y'(t) = -\mu_{ID}I(t) - \mu_{CD}C(t).$$

Since μ_{ID} , μ_{CD} , $S(t)$, $I(t)$, $C(t)$, and $R(t)$ are positive, it holds that

$$Y'(t) \leq Y(t),$$

and, due to the Comparison Principle 4.7, we obtain that

$$Y(t) \leq Y(0)e^t.$$

The last estimate assures us of the global existence of the solution.

We can also note that $Y'(t) \leq 0$, because of the positiveness of μ_{ID} , μ_{CD} , $I(t)$, and $C(t)$. It means that $Y(t)$ will decrease and its peak will be reached in $t = 0$. So, necessarily, it holds that

$$Y(t) \leq Y(0).$$

Consequentially, there exists a time-dependent function $D(t)$ such that

$$Y(t) = Y(0) + D(t).$$

This is how we define

$$D(t) := Y(0) - Y(t),$$

that is the number of people who died from the disease. The total population in the region will be denoted by

$$P(t) := Y(t) + D(t),$$

thereby $D(t) = P(0) - Y(t)$.

By removing the differential equation for deaths, we simplify the system of equations without losing information about mortality, which can be computed directly. The definition of $D(t)$ is intuitive: it simply represents the difference between the initial population and those still present in the compartments S, I, C, R . The equation $D(t) = P(0) - Y(t)$ ensures that the number of deaths aligns with the evolution of the other variables in the model.

Systems of differential equations, such as (1.1) and (1.2), can be applied to model the evolution of a disease through an entire population. However, these models do not make any distinction between rural and urban areas which can have widely differing population densities and which may in turn affect how quickly the disease is transmitted. A disease that is spread by person-to-person contact will spread much more quickly in an urban area as each individual will come into close contact with many more other people than someone who is living in a sparsely populated rural area. To consider this aspect we will propose a new model which describes the spread of a disease through different geographical regions of a country.

1.2.2 Equilibrium points

Now we can continue the qualitative analysis of the differential equations system (1.2) by finding its equilibrium points.

$$\begin{aligned} \frac{dS}{dt} &= -\lambda S(I + C) \\ \frac{dI}{dt} &= \lambda \beta S(I + C) - \mu_{IR}I - \mu_{ID}I \\ \frac{dC}{dt} &= \lambda(1 - \beta)S(I + C) - \mu_{CR}C - \mu_{CD}C \\ \frac{dR}{dt} &= \mu_{IR}I + \mu_{CR}C \end{aligned}$$

We have to pose the right-hand side of the system equal to zero. So from the first equation we obtain, since $\lambda \neq 0$, that either $S = 0$ or $I + C = 0$. As we are looking for positive solutions, the condition $I + C = 0$ implies necessarily that $I = C = 0$. Hence we distinguish two cases:

- If $S = 0$ from the second and the third equations we deduce that also I and C are equal to zero. By substituting I and C in the last equation, we infer that $\frac{dR}{dt} = 0$ and so $R(t) = R(0) = R_0$. We got a surface of equilibrium points $(0, 0, 0, R_0)$ with $R_0 > 0$.
- If $I = C = 0$, from the first equation we have that $\frac{dS}{dt} = 0$, so $S(t) = S(0) = S_1$, $\frac{dR}{dt} = 0$ and so $R(t) = R(0) = R_1$. Therefore the stationary points have the following form $(S_1, 0, 0, R_1)$, where S_1 and R_1 are greater than 0.

These stationary points reflect the end of an epidemic, where the disease has either been eradicated or has reached a point where it no longer spreads. The population is divided into those who were never infected (remaining susceptible), those who recovered, and those who died. In summary, these stationary points are realistic as they describe the final state of the population after the epidemic has run its course. Reaching this equilibrium point depends on many factors: effectiveness of interventions, disease transmission dynamics, and population behavior. Although it is an ideal state, it may not always be achievable without significant effort. If the disease can be controlled or eradicated through interventions such as vaccination, quarantine, or other measures, the population could theoretically reach this equilibrium.

Now let us study the stability of the equilibrium points by analyzing the Jacobian matrix. Let $u = (S, I, C, R)$ and $\dot{u} = F(u)$. Since the Jacobian matrix is

$$JF = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \cdots & \frac{\partial F_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \frac{\partial F_m}{\partial x_2} & \cdots & \frac{\partial F_m}{\partial x_n} \end{pmatrix},$$

in our case we have

$$JF = \begin{pmatrix} -\lambda(I + C) & -\lambda S & -\lambda S & 0 \\ \lambda\beta(I + C) & \lambda\beta S - (\mu_{IR} + \mu_{ID}) & \lambda\beta S & 0 \\ \lambda(1 - \beta)(I + C) & \lambda(1 - \beta)S & \lambda(1 - \beta)S - (\mu_{CR} + \mu_{CD}) & 0 \\ 0 & \mu_{IR} & \mu_{CR} & 0 \end{pmatrix}.$$

We now evaluate the Jacobian at equilibrium points, because we would like to check if they are hyperbolic to apply the Theorem 4.16. At first let us consider

$P_0 = (0, 0, 0, R_0)$, so we have

$$JF(P_0) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -(\mu_{IR} + \mu_{ID}) & 0 & 0 \\ 0 & 0 & -(\mu_{CR} + \mu_{CD}) & 0 \\ 0 & \mu_{IR} & \mu_{CR} & 0 \end{pmatrix},$$

which is a triangular matrix. It has 0 as eigenvalue with multiplicity 2 and the other eigenvalues are $-(\mu_{IR} + \mu_{ID})$ and $-(\mu_{CR} + \mu_{CD})$. They are negative, so we can expect these stationary points to be stable, but we have to find a different way to state that. We can try by searching for a Lyapunov function. Let $V(S, I, C, R) = S + I + C$. We are interested in positive solutions of the system (1.2), therefore we can consider a neighborhood U of P_0 such that $U = \{(S, I, C, R) \in \mathbb{R}^4 \mid (S, I, C) \neq (0, 0, 0) \wedge S, I, C \geq 0\}$. We need to check whether V satisfies the properties of a Lyapunov function for the equilibrium point P_0 to apply the Theorem 4.23:

- $V(P_0) = 0$.
- $V(P) > 0$ for every $P \in U \setminus \{P_0\}$.
- $\dot{V}(u) = \nabla V(u(t)) \cdot \dot{u}(t) = \nabla V(u) \cdot F(u) \leq 0$.

The first two conditions are fulfilled because of the choice of the neighborhood U . Now, we have to compute the orbital derivative of V .

$$\begin{aligned} \dot{V}(u) &= \nabla V(u) \cdot F(u) = -\lambda S(I + C) + \lambda \beta S(I + C) + \\ &\quad -\mu_{IR}I - \mu_{ID}I + \lambda(1 - \beta)S(I + C) - \mu_{CR}C - \mu_{CD}C \\ &= -(\mu_{IR} + \mu_{ID})I - (\mu_{CR} + \mu_{CD})C \leq 0. \end{aligned}$$

Hence V is a Lyapunov function for the equilibrium points like P_0 , which, in conclusion, are (Lyapunov) stable.

Secondly, we consider $P_1 = (S_1, 0, 0, R_1)$ and evaluate the Jacobian. We obtain

$$JF(P_1) = \begin{pmatrix} 0 & -\lambda S_1 & -\lambda S_1 & 0 \\ 0 & \lambda \beta S_1 - (\mu_{IR} + \mu_{ID}) & \lambda \beta S_1 & 0 \\ 0 & \lambda(1 - \beta)S_1 & \lambda(1 - \beta)S_1 - (\mu_{CR} + \mu_{CD}) & 0 \\ 0 & \mu_{IR} & \mu_{CR} & 0 \end{pmatrix}.$$

It has 0 as an eigenvalue with multiplicity 2 and the other two eigenvalues are the roots of the determinant of the matrix

$$\begin{pmatrix} \lambda \beta S_1 - (\mu_{IR} + \mu_{ID}) - z & \lambda \beta S_1 \\ \lambda(1 - \beta)S_1 & \lambda(1 - \beta)S_1 - (\mu_{CR} + \mu_{CD}) - z \end{pmatrix}.$$

The determinant is

$$z^2 + (\mu_{IR} + \mu_{ID} + \mu_{CR} + \mu_{CD} - \lambda S_1)z + (\mu_{IR} + \mu_{ID})(\mu_{CR} + \mu_{CD}) - \lambda S_1[\beta(\mu_{CR} + \mu_{CD}) + (1 - \beta)(\mu_{IR} + \mu_{ID})].$$

Now we define

- $c = \mu_{IR} + \mu_{ID} + \mu_{CR} + \mu_{CD}$.
- $p = (\mu_{IR} + \mu_{ID})(\mu_{CR} + \mu_{CD})$.
- $c_\beta = \beta(\mu_{CR} + \mu_{CD}) + (1 - \beta)(\mu_{IR} + \mu_{ID})$.

Therefore the roots are

$$z_{1,2} = \frac{-(c - \lambda S_1) \pm \sqrt{(c - \lambda S_1)^2 - 4(p - \lambda S_1 c_\beta)}}{2}.$$

The sign of the real part of the roots depends on the factor λS_1 . Therefore the real part could be positive or negative.

1.3 Introduction of vaccination

The disease can be controlled or eradicated through interventions such as vaccination, quarantine, or other measures.

Vaccination changes the course of disease spread. By administering vaccines to a significant portion of the population, we reduce the number of susceptible individuals who can be infected. Essentially, vaccination shifts people from the susceptible (S) category to what we can call an immune (R) or vaccinated (V) category without them ever getting sick. In a simplified version of this model, if we vaccinate enough people, the virus has fewer targets to infect, which drastically reduces its spread. This concept is called herd immunity, where a sufficient number of people are immune to a disease, either through vaccination or previous infection, and the disease struggles to find new hosts.

The following is an example of a SIRCD model with the presence of vaccination

$$\begin{aligned} \frac{dS}{dt} &= -\lambda S(I + C) - \nu S \\ \frac{dI}{dt} &= \lambda \beta S(I + C) - \mu_{IR}I - \mu_{ID}I \\ \frac{dC}{dt} &= \lambda(1 - \beta)S(I + C) - \mu_{CR}C - \mu_{CD}C \\ \frac{dR}{dt} &= \mu_{IR}I + \mu_{CR}C + \nu S \end{aligned} \tag{1.3}$$

where ν is the vaccination rate, which we suppose to be constant and, obviously greater than 0, at the moment. This change in the equations has a profound effect on the overall dynamics. With a high enough ν , the susceptible population shrinks rapidly, and fewer people ever get infected. The key takeaway is that the larger the proportion of vaccinated individuals, the lower the chance of the disease continuing to spread.

1.3.1 Positiveness of the solution

Before studying the stability of the equilibrium points we can note that the solutions of the system (1.3) are non-negative for every t in the existence interval. The motivation is based on the same proof present in the section 1.2.1.

Assume $S(0) > 0$. We will prove by contradiction that $S(t) > 0$ for all $t \geq 0$. Suppose, for the sake of contradiction, that $S(t)$ becomes non-positive at some $t_1 > 0$, meaning $S(t_1) = 0$. From the differential equation for $S(t)$:

$$\frac{dS}{dt} = -\lambda S(I + C) - \nu S.$$

At t_1 , where $S(t_1) = 0$, we have:

$$\frac{dS}{dt} = 0.$$

Therefore, $S(t)$ reaches zero at t_1 with a zero derivative, meaning $S(t)$ cannot decrease further and remains zero after t_1 . For $t \in [0, t_1)$, we have $S(t) > 0$. In this region, the differential equation for $S(t)$ is:

$$\frac{dS}{dt} = -\lambda S(I + C) - \nu S.$$

Since $S(t) > 0$, it follows that $\frac{dS}{dt} \leq 0$, meaning $S(t)$ is non-increasing. This implies that $S(t)$ can only decrease or remain constant, and therefore cannot reach zero unless it decreases to zero. The assumption that $S(t)$ could become negative at some point leads to a contradiction because once $S(t)$ reaches zero, it cannot decrease further. Thus, $S(t)$ cannot become negative after t_1 . Hence, $S(t) \geq 0$ for all $t \geq 0$. Moreover, since $S(0) > 0$, we conclude that $S(t) > 0$ for all $t \geq 0$.

Now we prove that also $I(t)$ and $C(t)$ fulfill this property. If by contradiction $I(t) < 0$ for any t , then, due to the Sign Permanence Theorem and the continuity of I , there exists t_0 such that $I(t_0) = 0$ and, after t_0 , I is negative. Now we define

$$\Omega_I = \{t \in (0, t_0] \mid I(t) = 0, \exists \delta > 0 \text{ such that } I(s) < 0 \forall s \in (t, t + \delta]\}$$

and $t_m := \min \Omega_I$, that is well defined as Ω_I is not empty ($t_0 \in \Omega_I$), I is a continuous function and $I(0) > 0$. We note that

$$\dot{I}(t_m) = \lambda\beta S(t_m)C(t_m),$$

so the sign of the right-hand side is determined by that one of C , because of the positiveness of λ, β and S , previously showed. If $C(t_m) \geq 0$, then $\dot{I}(t_m) \geq 0$, therefore I will not decrease and this fact is in contradiction with $I(s) < 0$ for every $s \in (t_m, t_m + \delta]$. Instead, if $C(t_m) < 0$, then $\dot{I}(t_m) < 0$. At this point, we consider the equation of C . We define

$$\Omega_C = \{t \in (0, t_m] \mid C(t) = 0, \exists \delta > 0 \text{ such that } C(s) < 0 \forall s \in (t, t + \delta]\}$$

and $t_k := \min \Omega_C$. Since $t_k \leq t_m$, $I(s) \geq 0$ for every $s \in (0, t_k]$. So we get

$$\dot{C}(t_k) = \lambda(1 - \beta)S(t_k)I(t_k) \geq 0,$$

that is in contradiction with the definition of t_k , as the non-negative derivative implies the non-degrowth of the function. In conclusion, I will remain positive, if the initial condition $I(0) > 0$. By applying the same argument, we can prove the positiveness of C .

In conclusion, we have to prove the positiveness of R , by knowing that S , I , C , and $R(0)$ are positive. Since

$$\dot{R}(t) = \mu_{IR}I(t) + \mu_{CR}C(t) + \nu S(t).$$

it holds that the derivative of R is non-negative. Now $R(0)$ is positive, therefore R will not decrease and will remain positive for every t in the existence interval.

1.3.2 Qualitative analysis of the model

We can make a qualitative analysis by finding the equilibrium points of the system 1.3. If we pose the right-hand side of the system equal to zero, we obtain from the fourth equation that $S = I = C = 0$ as we are looking for positive solutions. By substituting S , I , and C in the last two equations, we infer that $\frac{dR}{dt} = 0$ and so $R(t) = R(0) = R_0$. We got a surface of equilibrium points $(0, 0, 0, R_0)$ with $R_0 > 0$.

By considering $V(S, I, C, R) = S + I + C$ we can deduct the stability of the equilibrium points. Let W be a neighborhood of P_0 , one of the stationary points, such that $W = \{(S, I, C, R) \in \mathbb{R}^4 \mid (S, I, C) \neq 0 \wedge S, I, C \geq 0\}$. We need to check whether V satisfies the properties of a Lyapunov function for the equilibrium point P_0 to apply the Theorem 4.23:

- $V(P_0) = 0$.

- $V(P) > 0$ for every $P \in W \setminus \{P_0\}$.
- $\dot{V}(u) = \nabla V(u(t)) \cdot \dot{u}(t) = \nabla V(u) \cdot F(u) \leq 0$.

The first two conditions are satisfied because of the choice of the neighborhood W . Now we have to compute the orbital derivative of V . We have that

$$\begin{aligned} \dot{V}(u) &= \nabla V(u) \cdot F(u) = -\lambda S(I + C) - \nu S + \lambda \beta S(I + C) + \\ &\quad - \mu_{IR}I - \mu_{ID}I + \lambda(1 - \beta)S(I + C) - \mu_{CR}C - \mu_{CD}C \\ &= -\nu S - (\mu_{IR} + \mu_{ID})I - (\mu_{CR} + \mu_{CD})C \leq 0. \end{aligned}$$

Hence V is a Lyapunov function for the equilibrium points like P_0 , which, in conclusion, are (Lyapunov) stable.

Now, if we define $Y(t) = S(t) + I(t) + C(t) + R(t)$, then we get, like previously in the final part of the section 1.1, the global existence of the solution.

We can also note that

$$Y'(t) = -\mu_{ID}I - \mu_{CD}C \leq 0,$$

because of the positiveness of μ_{ID} , μ_{CD} , $I(t)$, and $C(t)$. It means that $Y(t)$ will decrease and its peak will be reached in $t = 0$. So, necessarily, it holds that

$$Y(t) \leq Y(0).$$

Consequentially, there exists a time-dependent function $D(t)$ such that

$$Y(t) = Y(0) + D(t).$$

This is how we define

$$D(t) := Y(0) - Y(t),$$

that is the number of people who died from the disease. The total population in the region will be denoted by

$$P(t) := Y(t) + D(t),$$

thereby $D(t) = P(0) - Y(t)$.

In conclusion, we want to say something about the behavior of the compartments of recovered people. We note immediately that:

$$S(t) + I(t) + C(t) \leq S(t) + I(t) + C(t) + R(t) = Y(t) \leq Y(0).$$

Since

$$\frac{dR}{dt} = \mu_{IR}I(t) + \mu_{CR}C(t) + \nu S(t),$$

if we integrate, we get that

$$\begin{aligned} R(t) &= R(0) + \int_0^t \mu_{IR}I(\tau) + \mu_{CR}C(\tau) + \nu S(\tau) d\tau \\ &\leq R(0) + (\mu_{IR} + \mu_{CR} + \nu) \int_0^t Y(0) d\tau \\ &= R(0) + (\mu_{IR} + \mu_{CR} + \nu)Y(0)t, \end{aligned}$$

that is, the class of recovered people grows at most linearly.

A benefit of the introduction of vaccination is a marked improvement in the behavior of susceptibles. Because of the following inequality

$$\frac{dS}{dt} = -\lambda S(t)(I(t) + C(t)) - \nu S(t) \leq -\nu S(t),$$

our goal is to apply 4.7 by considering the equation $\frac{d\tilde{S}}{dt} = \nu\tilde{S}$ with the initial condition $\tilde{S}(0)$, such that $\tilde{S}(0) = S(0)$. By applying 4.7, we get that

$$S(t) \leq \tilde{S}(t) = S(0)e^{-\nu t}.$$

Hence $S(t) \rightarrow 0$ exponentially. In conclusion, we have asymptotic stability in the variable of susceptibles.

1.4 Possible improvements of the model

Improving the SIRCD (Susceptible-Infected-Recovered-Carriers-Deceased) model can involve several strategies to enhance its accuracy and applicability. We suggest two ways.

1.4.1 Time-Dependent Parameters Extension

In an epidemiological model, parameters such as the transmission rate λ or vaccination rate ν might change over time due to interventions like lockdowns, public health measures, or vaccination campaigns. Incorporating time-dependent parameters allows the model to account for these changes dynamically.

Let us modify the transmission rate, λ , and vaccination rate, ν , to be time-dependent functions: $\lambda(t)$ and $\nu(t)$.

$$\begin{aligned}
\frac{dS}{dt} &= -\lambda(t)S(I + C) - \nu(t)S \\
\frac{dI}{dt} &= \lambda(t)\beta S(I + C) - \mu_{IR}I - \mu_{ID}I \\
\frac{dC}{dt} &= \lambda(t)(1 - \beta)S(I + C) - \mu_{CR}C - \mu_{CD}C \\
\frac{dR}{dt} &= \mu_{IR}I + \mu_{CR}C + \nu(t)S
\end{aligned} \tag{1.4}$$

The infection rate $\lambda(t)$ can be influenced by several factors, such as public health interventions, vaccination rates, or changes in behavior due to the outbreak. Let us see some possible functions to assign to $\lambda(t)$:

1. Constant: $\lambda(t) = \lambda_0$, where λ_0 is a constant. This assumes a constant rate of transmission over time.
2. Periodic: $\lambda(t) = \lambda_0(1 + \alpha \cos(\omega t))$, where α and ω are constants. This can model seasonal variations in transmission rates.
3. Exponential Decay: $\lambda(t) = \lambda_0 e^{-\delta t}$, where δ is a decay rate. This can represent the effect of interventions reducing transmission over time and model rapid increases in infection rates, particularly at the beginning of an outbreak.

In general, $\nu(t)$ plays a critical role in determining how susceptible individuals are converted to a vaccinated status, thereby reducing the number of individuals susceptible to infection. When modeling the vaccination rate $\nu(t)$, we have to consider the dynamics of vaccination campaigns, population responses, and other sociocultural factors. Let us see some possible examples of $\nu(t)$:

1. Constant: $\nu(t) = \nu_0$, where ν_0 is a constant. It represents a steady vaccination rate over time, which may be appropriate in stable contexts where vaccination programs are consistently applied. This assumes a constant rate of natural death or recovery.
2. Linear: $\nu(t) = \nu_0 + \nu_1 t$, where ν_0 and ν_1 are constants. It represents a gradually increasing or decreasing vaccination rate, which could reflect the scaling up of vaccination efforts over time.
3. Exponential decay: $\nu(t) = \nu_0(1 - e^{-kt})$, where k is a decay rate. This model reflects a rapid increase in vaccination rates at first, which then levels off as the program reaches saturation.
4. Logistic Growth: $\nu(t) = \frac{\nu_0}{1 + e^{-\kappa(t-t_0)}}$, where ν_0 and t_0 are constants. This models a scenario where the vaccination rate starts slowly, increases rapidly, and then slows down as it approaches a maximum capacity ν_0 (the

maximum vaccination rate). This is particularly useful in representing real-world vaccination campaigns that face logistical challenges as they scale.

1.4.2 Age-Based Extensions

In reality, the risk of infection, recovery, and death can vary significantly by age group. By extending the SICRD model to incorporate different age groups, we can more accurately model the spread of the disease in the population. We will divide the population into different age-based compartments, such as young, adult, and elderly.

Let us extend the SICRD model by dividing the population into three age groups: young Y , adults A , and elderly E . Each group will have its own set of equations, but they will interact with each other through transmission dynamics. The differential equation systems are respectively for the Young, Adult, and Elderly Age Groups:

$$\begin{aligned}
\frac{dS_Y}{dt} &= -\lambda_Y S_Y (I_Y + I_A + I_E + C_Y + C_A + C_E) - \nu_Y S_Y \\
\frac{dI_Y}{dt} &= \lambda_Y \beta_Y S_Y (I_Y + I_A + I_E + C_Y + C_A + C_E) - (\mu_{IR_Y} + \mu_{ID_Y}) I_Y \\
\frac{dC_Y}{dt} &= \lambda_Y (1 - \beta_Y) S_Y (I_Y + I_A + I_E + C_Y + C_A + C_E) - (\mu_{CR_Y} + \mu_{CD_Y}) C_Y \\
\frac{dR_Y}{dt} &= \mu_{IR_Y} I_Y + \mu_{CR_Y} C_Y + \nu_Y S_Y
\end{aligned} \tag{1.5}$$

$$\begin{aligned}
\frac{dS_A}{dt} &= -\lambda_A S_A (I_Y + I_A + I_E + C_Y + C_A + C_E) - \nu_A S_A \\
\frac{dI_A}{dt} &= \lambda_A \beta_A S_A (I_Y + I_A + I_E + C_Y + C_A + C_E) - (\mu_{IR_A} + \mu_{ID_A}) I_A \\
\frac{dC_A}{dt} &= \lambda_A (1 - \beta_A) S_A (I_Y + I_A + I_E + C_Y + C_A + C_E) - (\mu_{CR_A} + \mu_{CD_A}) C_A \\
\frac{dR_A}{dt} &= \mu_{IR_A} I_A + \mu_{CR_A} C_A + \nu_A S_A
\end{aligned} \tag{1.6}$$

$$\begin{aligned}
\frac{dS_E}{dt} &= -\lambda_E S_E (I_Y + I_A + I_E + C_Y + C_A + C_E) - \nu_E S_E \\
\frac{dI_E}{dt} &= \lambda_E \beta_E S_E (I_Y + I_A + I_E + C_Y + C_A + C_E) - (\mu_{IR_E} + \mu_{ID_E}) I_E \\
\frac{dC_E}{dt} &= \lambda_E (1 - \beta_E) S_E (I_Y + I_A + I_E + C_Y + C_A + C_E) - (\mu_{CR_E} + \mu_{CD_E}) C_E \\
\frac{dR_E}{dt} &= \mu_{IR_E} I_E + \mu_{CR_E} C_E + \nu_E S_E
\end{aligned} \tag{1.7}$$

In all compartments, the transmission depends on the total number of infected individuals from all age groups and carriers. This means that young, adult, and elderly individuals can all contribute to the infection of others, regardless of their age. The parameter λ varies by age group to reflect differences in contact rates or susceptibility.

By dividing the population into age-based compartments, the model becomes more detailed and can capture the heterogeneity in disease dynamics across different age groups. This is crucial when modeling diseases like COVID-19, where the elderly are more vulnerable to severe outcomes, while younger populations might have higher contact rates but lower mortality.

Introduction of diffusion

In this chapter, we will introduce a diffusion factor in our model (1.2). Diseases spread not only based on individual interactions but also across geographical regions. In reality, people are not uniformly mixed across space, and the spread of disease is influenced by population density, mobility, and location-specific factors.

2.1 Regional kinetics model

Incorporating geography into the model could be really meaningful since it allows us to account for localized outbreaks and mobility patterns. Some areas might experience more intense outbreaks due to higher population densities or higher contact rates (e.g., urban vs. rural areas). People travel between regions (commuting, migration, tourism, etc.), which can introduce the infection to new areas or cause secondary outbreaks. Modeling movement between geographical areas can help capture this spread more accurately. Introducing a geographical factor into the SIRCD model is crucial for improving the accuracy and realism of predictions. It allows us to capture the spatial heterogeneity of disease spread, account for regional differences in population density, healthcare capacity, and public policies, and model how mobility and travel influence the dynamics. By incorporating geography, we can better simulate localized interventions, prepare for future waves of infection, and ultimately design more effective, region-specific public health policies.

A large population may be divided into a number of smaller populations according to the geographical zone in which the people live. In this case, it is likely that the rates at which people recover or die from a disease are the same in every region but the infection rate λ may be different in each area and may

depend on quantities such as population density. The model will also need to include the effect of people moving between the different zones. Here we will assume that the movement of the people does not produce any change in the population of any one territory. That is, the number of people who leave region Ω will be the same as those who enter region Ω . Assume that the country or state under consideration is divided into N geographical regions, where it is also assumed that if the country is divided into small enough area any inhomogeneities in the population distribution within each zone can be neglected.

Let S_i , I_i , C_i , and R_i denote the number of people who are susceptible, infected, carriers, or who recovered in the i th region respectively; and let $\mathbf{S}$, $\mathbf{I}$, $\mathbf{C}$, and $\mathbf{R}$ denote the vectors which list the values of the corresponding category in each region. For simplicity of notation, introduce the vector-valued function $\mathbf{F}$ defined by $F_i = S_i(I_i + C_i)$. The differential equations (1.2) for a single region can be extended to the multi-region case and, like in the Article [5, Section 2.2], can be expressed as

$$\begin{aligned}\frac{d\mathbf{S}}{dt} &= -\lambda\mathbf{F} + T\mathbf{S} \\ \frac{d\mathbf{I}}{dt} &= \lambda\beta\mathbf{F} - \mu_{IR}\mathbf{I} - \mu_{ID}\mathbf{I} + T\mathbf{I} \\ \frac{d\mathbf{C}}{dt} &= \lambda(1 - \beta)\mathbf{F} - \mu_{CR}\mathbf{C} - \mu_{CD}\mathbf{C} + T\mathbf{C} \\ \frac{d\mathbf{R}}{dt} &= \mu_{IR}\mathbf{I} + \mu_{CR}\mathbf{C} + T\mathbf{R}\end{aligned}\tag{2.1}$$

where T is a matrix $N \times N$ which models how the population moves between the different regions. In the model presented here the matrix T is constructed so that the total population of living people in each region remains the same, and the total number of people in each category is not changed by the transport terms. Let us also assume that the matrix T is negative definite.

2.1.1 Qualitative analysis of the multi-regional model

Now, if we define $\mathbf{Y}(t) = \mathbf{S}(t) + \mathbf{I}(t) + \mathbf{C}(t) + \mathbf{R}(t)$, then we get

$$\mathbf{Y}'(t) = -\mu_{ID}\mathbf{I}(t) - \mu_{CD}\mathbf{C}(t) + T[\mathbf{S}(t) + \mathbf{I}(t) + \mathbf{C}(t) + \mathbf{R}(t)].$$

Since μ_{ID} , μ_{CD} , $\mathbf{I}(t)$, and $\mathbf{C}(t)$ are positive (this, for the moment, is taken as given, but in general, it can be proven under suitable assumptions), it holds that

$$\mathbf{Y}'(t) \leq T\mathbf{Y}(t),$$

where $\leq$ is intended component by component. Now, due to the Comparison Principle 4.7, we get that

$$\mathbf{Y}(t) \leq e^{Tt} \mathbf{Y}(0).$$

Since T is negative definite, e^{Tt} is a contraction, that is $\|e^{Tt}\| \leq 1$, then we obtain $\mathbf{Y}(t) \leq \mathbf{Y}(0)$, therefore $Y_i(t) \leq Y_i(0)$ for every $i = 0, \dots, N$. Consequentially, there exists a time-dependent function $D_i(t)$ such that $Y_i(t) = Y_i(0) + D_i(t)$. This is how we define

$$D_i(t) := Y_i(0) - Y_i(t),$$

that is the number of people who died from the disease in the i th region. Accordingly, $\mathbf{D}(t)$ will be the vector that lists the values of the category in each region. The total population in the i th region will be denoted by

$$P_i(t) := Y_i(t) + D_i(t),$$

thereby $D_i(t) = P_i(0) - Y_i(t)$.

In general, we can make the model compact, that is

$$\begin{pmatrix} \dot{\mathbf{S}} \\ \dot{\mathbf{I}} \\ \dot{\mathbf{C}} \\ \dot{\mathbf{R}} \end{pmatrix} + A \begin{pmatrix} \mathbf{S} \\ \mathbf{I} \\ \mathbf{C} \\ \mathbf{R} \end{pmatrix} + L \begin{pmatrix} \mathbf{S} \\ \mathbf{I} \\ \mathbf{C} \\ \mathbf{R} \end{pmatrix} = G \begin{pmatrix} \mathbf{S} \\ \mathbf{I} \\ \mathbf{C} \\ \mathbf{R} \end{pmatrix}.$$

Let $u = (\mathbf{S}, \mathbf{I}, \mathbf{C}, \mathbf{R})$, so we have

$$\dot{u} + \tilde{A}u + Lu = G(u),$$

where $\tilde{A}$ and L are matrices $4N \times 4N$. They describe the linear part, while the function G encapsulates the nonlinear dynamics. According to the considerations made previously, the matrix $\tilde{A}$ and L will have the following form:

$$\tilde{A} = \begin{pmatrix} -T & 0 & \dots & 0 \\ 0 & -T & \ddots & \vdots \\ \vdots & \ddots & -T & 0 \\ 0 & \dots & 0 & -T \end{pmatrix}, \quad L = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \mu_{IR}I & 0 & 0 \\ 0 & 0 & \mu_{ID}I & 0 \\ 0 & -\mu_{IR}I & -\mu_{CR}I & 0 \end{pmatrix}$$

where T , 0 , and I are matrices $N \times N$ and I is the identity matrix.

Now we incorporate the entire linear part into a matrix, which we denote by A . Therefore we get:

$$\dot{u} + Au = G(u). \tag{2.2}$$

So we obtain that the general solution of the system is given by

$$u(t) = e^{-At}u_0 + \int_0^t e^{-A(t-\tau)}G(u(\tau)) d\tau ,$$

where $u_0 = (\mathbf{S}(0), \mathbf{I}(0), \mathbf{C}(0), \mathbf{R}(0)) = (\mathbf{S}_0, \mathbf{I}_0, \mathbf{C}_0, \mathbf{R}_0)$.

Remark 2.1. Under suitable assumptions on the matrix A , the positiveness of the solution can be shown if we start from positive initial conditions. One of the most typical assumptions for A is that it is an M-Matrix. A proof of this property can be found in the text [11, Theorem 6.5].

The class of Z-matrices are those matrices whose off-diagonal entries are less than or equal to zero; that is, the matrices of the form:

$$Z = (z_{ij}), \quad z_{ij} \leq 0, \quad i \neq j.$$

Definition 2.2. Let A be a $n \times n$ real Z-matrix. Then matrix A is also an M-matrix if it can be expressed in the form $A = sI - B$, where $B = (b_{ij})$ with $b_{ij} > 0$, for all $1 \leq i, j \leq n$, where s is greater than the maximum of the moduli of the eigenvalues of B , and I is an identity matrix.

We can assume that A is a diagonal matrix. What we want to prove is the following property:

Lemma 2.3. Let $x \in \mathbb{C}^n$, $A \in \mathcal{M}(n, \mathbb{C})$ a normal matrix, that is $A^*A = AA^*$, and $G : \mathbb{C}^n \rightarrow \mathbb{C}^n$ a vector function. Let us suppose to have the system

$$\dot{x} + Ax = G(x). \tag{2.3}$$

Then we can reduce it to

$$\dot{y} + Dy = \tilde{G}(y), \tag{2.4}$$

where $y \in \mathbb{C}^n$, $D \in \mathcal{M}(n, \mathbb{C})$ is a diagonal matrix and $\tilde{G} : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is a vector function.

Proof. The idea is to use the Spectral Theorem and make a change of basis. Due to 4.28, A is unitarily similar to a diagonal matrix D , whose diagonal elements are the eigenvalues of A . Therefore there exists $U \in \mathcal{M}(n, \mathbb{C})$, unitary matrix, such that $U^{-1}AU = U^*AU = D$. Since U is unitary, for every $x \in \mathbb{C}^n$ there exists a unique $y \in \mathbb{C}^n$ such that $Uy = x$. Thus we have $U\dot{y} = \dot{x}$ and, by substituting in the initial equation (2.3), we get

$$\dot{x} + AUy = G(Uy) = (GU)y.$$

Now, by applying U^{-1} on both sides, we obtain

$$U^{-1}\dot{x} + U^{-1}AUy = U^{-1}(GU)y,$$

that is,

$$\dot{y} + Dy = (U^{-1}GU)y = \tilde{G}y,$$

hence the claim follows. $\square$

2.2 Global existence of the solution

The aim of this section is to prove the global existence of the solution of the system (2.1). We will need some additional assumptions and a preliminary lemma. We consider the system in the compact version like in (2.2), that is

$$\dot{u}(t) + Au(t) = G(u(t)),$$

where we assume that $A \in \mathcal{M}(4N, \mathbb{C})$ is a self-adjoint and positive definite matrix. The last two properties can be expressed also in this way: the eigenvalues λ_j of A are positive real numbers. We indicate by $Sp(A)$ the set of the eigenvalues of A .

Now we define

$$B_T^n = \{u : [0, T] \rightarrow \mathbb{C}^n \mid u \in C^0([0, T])\}.$$

Therefore, our first step is to prove the following

Lemma 2.4. *Let $A \in \mathcal{M}(n, \mathbb{C})$ be a self-adjoint and positive definite matrix. Then it holds the estimate*

$$\|e^{-At}f\| \leq e^{-\theta t}\|f\|,$$

where $\theta = \inf_{\lambda \in Sp(A)} \lambda$, $f \in B_T^n$, and $\|\cdot\|$ is the euclidean norm on $\mathbb{C}^n$.

Proof. Since A is a self-adjoint and positive definite matrix, thanks to the Spectral Theorem 4.28 there exists a unitary matrix $U \in \mathcal{M}(n, \mathbb{C})$ such that $U^{-1}AU = U^*AU = D$, where $D \in \mathcal{M}(n, \mathbb{C})$ is a diagonal matrix whose diagonal elements are the eigenvalues $\lambda_j > 0$ of A . So, due to a property of matrix exponential, we have

$$e^{-At} = e^{-UDU^*t} = Ue^{-Dt}U^*.$$

The Euclidean norm is invariant under unitary transformation, therefore we get

$$\|e^{-At}f\| = \|Ue^{-Dt}U^*f\| = \|e^{-Dt}f\|.$$

The exponential of a diagonal matrix is the diagonal matrix with the exponential of the elements, that is

$$e^{-Dt} = \begin{pmatrix} e^{-\lambda_1 t} & 0 & \dots & 0 \\ 0 & e^{-\lambda_2 t} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & e^{-\lambda_n t} \end{pmatrix},$$

hence

$$e^{-Dt}f = \begin{pmatrix} e^{-\lambda_1 t}f_1 \\ e^{-\lambda_2 t}f_2 \\ \vdots \\ e^{-\lambda_n t}f_n \end{pmatrix}.$$

If we define $\theta := \inf_{\lambda \in Sp(A)} \lambda$, we obtain that $e^{-\lambda_j t}f_j \leq e^{-\theta t}f_j$ for every $j = 1, \dots, n$, so the claim follows, that is $\|e^{-At}f\| = \|e^{-Dt}f\| \leq e^{-\theta t}\|f\|$. $\square$

Remark 2.5. In the previous lemma if we use a different norm, the claim is always true because of the equivalence of the norms, but it appears a constant due to the scaling.

Eventually, we are ready to state and prove the main Theorem that guarantees us the global existence and the uniqueness of the solution of the Cauchy Problem

$$\begin{cases} \dot{u}(t) + Au(t) = G(u(t)) & u \in B_T^{4N} \\ u(0) = u_0 & u_0 \in \mathbb{C}^{4N}, \end{cases} \quad (2.5)$$

where u_0 is fixed.

Theorem 2.6 (Global Existence and Uniqueness Theorem). *Let us consider the Cauchy Problem (2.5) and let $\|\cdot\|$ be the Euclidean norm. We assume that:*

1. *G is a locally Lipschitz continuous function with respect to u , that is for every $R > 0$ there exists a constant $C(R) > 0$ such that if $\|f_1\|, \|f_2\| \leq R$, it holds that $\|G(f_1) - G(f_2)\| \leq C(R)\|f_1 - f_2\|$;*
2. *$G(0) = 0$ and the existence of $c, \sigma \in \mathbb{R}^+$ such that $C(R) \leq cR^\sigma$ (hence $C(R) \rightarrow 0^+$ as $R \rightarrow 0^+$);*

3. $A \in \mathcal{M}(4N, \mathbb{C})$ is a self-adjoint and positive definite matrix.

Then there exists $\varepsilon_0 \in \mathbb{R}^+$ such that if $\|u_0\| \leq \varepsilon_0$, there exists a unique solution $u \in \mathcal{C}^0([0, +\infty); \mathbb{C}^{4N})$ of the Cauchy Problem (2.5).

Proof. At first, we have to rewrite the Cauchy Problem (2.5) in the integral version. Through a series of algebraic manipulations, it is possible to say that $u \in \mathcal{C}^0([0, +\infty); \mathbb{C}^{4N})$ is a solution of (2.5) if and only if

$$u(t) = e^{-At}u_0 + \int_0^t e^{-A(t-\tau)}G(u(\tau)) d\tau.$$

Now we define the operator

$$K(u(t)) := e^{-At}u_0 + \int_0^t e^{-A(t-\tau)}G(u(\tau)) d\tau,$$

and, given $T \in \mathbb{R}^+$,

$$B_T := B_T^{4N} = \{u : [0, T] \rightarrow \mathbb{C}^{4N} \mid u \in \mathcal{C}^0([0, T])\}.$$

We consider on B_T the norm

$$\|u\|_{B_T} := \sup_{t \in [0, T]} e^{\theta t} \|u(t)\|,$$

where $\theta = \inf_{\lambda \in Sp(A)} \lambda$. So $(B_T, \|\cdot\|_{B_T})$ is a Banach space. Our aim is to prove the existence and uniqueness of the solution on the interval $[0, T]$ thanks to the Banach fixed-point Theorem 4.32. The proof will be divided into two steps:

1. to prove that $K : B_{B_T}(\delta) \longrightarrow B_{B_T}(\delta)$ is a well defined function, where δ has to be chosen and $B_{B_T}(\delta) = \{u \in B_T \mid \|u\|_{B_T} \leq \delta\}$ is closed in B_T , so it is a complete metric space as well.
2. To prove that K is a contraction.

First of all, we show that K is a well defined function. Let $\delta = 2\varepsilon_0$, where the existence of ε_0 has yet to be ascertained, but by the end of this point it will be clear. The claim is that if $u \in B_{B_T}(\delta)$, then $K(u) \in B_{B_T}(\delta)$. By starting from the definition of K , we get that

$$\|K(u(t))\| = \left\| e^{-At}u_0 + \int_0^t e^{-A(t-\tau)}G(u(\tau)) d\tau \right\|,$$

and, by using 2.4 and triangular inequality, we obtain

$$\|K(u(t))\| \leq e^{-\theta t} \|u_0\| + \int_0^t e^{-\theta(t-\tau)} \|G(u(\tau))\| d\tau.$$

Since u is a continuous function on a compact interval there exists a real number $R \in \mathbb{R}^+$ such that $\|u\| \leq R$. Corresponding to this value, thanks to the assumptions on G , there exists a constant $C(R) \in \mathbb{R}^+$ such that for every $\tau \in [0, t]$ it holds that $\|G(u(\tau))\| \leq C(R)\|u(\tau)\|$. By applying this inequality, we obtain

$$\|K(u(t))\| \leq e^{-\theta t}\|u_0\| + \int_0^t e^{-\theta(t-\tau)}C(R)\|u(\tau)\| d\tau.$$

Now, due to the definition of $\|\cdot\|_{B_T}$, it holds that

$$\|u(\tau)\| \leq e^{-\theta\tau}\|u\|_{B_T} \leq \|u\|_{B_T} \leq \delta =: R.$$

By using this inequality, the facts that $C(R) \leq cR^\sigma = c\delta^\sigma$ and $\|u_0\| \leq \varepsilon_0$, we gain that

$$\begin{aligned} \|K(u(t))\| &\leq e^{-\theta t}\varepsilon_0 + c \int_0^t e^{-\theta(t-\tau)}\delta^{1+\sigma} d\tau \\ &\leq e^{-\theta t}\varepsilon_0 + ce^{-\theta t}\delta^{1+\sigma} \int_0^t e^{\theta\tau} d\tau. \end{aligned}$$

By multiplying by $e^{\theta t}$ in both sides, we have

$$e^{\theta t}\|K(u(t))\| \leq \varepsilon_0 + c\delta^{1+\sigma} \int_0^t e^{\theta\tau} d\tau = \varepsilon_0 + c\delta^{1+\sigma}e^{\theta T}.$$

Now if we take the supremum on the interval $[0, T]$, we get

$$\|K(u)\|_{B_T} := \sup_{t \in [0, T]} e^{\theta t}\|K(u(t))\| \leq \varepsilon_0 + c\delta^{1+\sigma}e^{\theta T}.$$

Since $u \in B_{B_T}(\delta)$ and $\delta = 2\varepsilon_0$, we have

$$\|K(u)\|_{B_T} \leq \varepsilon_0 + c(2\varepsilon_0)^{1+\sigma}e^{\theta T}.$$

Our aim is to show that $K(u) \in B_{B_T}(2\varepsilon_0)$, therefore we impose that

$$\varepsilon_0 + c(2\varepsilon_0)^{1+\sigma}e^{\theta T} \leq 2\varepsilon_0,$$

by obtaining that

$$\varepsilon_0 \leq \frac{1}{2(ce^{\theta T})^{\frac{1}{\sigma}}}.$$

Eventually, by choosing ε_0 in this way and by taking $\delta = 2\varepsilon_0$, we get that $K : B_{B_T}(\delta) \rightarrow B_{B_T}(\delta)$ is a well defined function.

The second part consists in proving that K is a contraction. The proof will be similar to the previous one. Let $u_1, u_2 \in B_{B_T}(2\varepsilon_0)$ and $t \in [0, T]$. We have

$$\|K(u_1(t)) - K(u_2(t))\| \leq \left\| \int_0^t e^{-A(t-\tau)} [G(u_1(\tau)) - G(u_2(\tau))] d\tau \right\|.$$

Due to the lemma 2.4 and a property of the integral, it holds that

$$\|K(u_1(t)) - K(u_2(t))\| \leq \int_0^t e^{-\theta(t-\tau)} \|G(u_1(\tau)) - G(u_2(\tau))\| d\tau.$$

Our intention is to use the fact that G is a local Lipschitz function. Since $u_1, u_2 \in B_{B_T}(2\varepsilon_0)$, for $i = 1, 2$ it holds that

$$\|u_i\|_{B_T} := \sup_{t \in [0, T]} e^{\theta t} \|u_i(t)\| \leq 2\varepsilon_0,$$

therefore for every $t \in [0, T]$ we have

$$\|u_i(t)\| \leq e^{-\theta t} \|u_i\|_{B_T} \leq 2\varepsilon_0 := R(\varepsilon_0).$$

Corresponding to $2\varepsilon_0$ there exist three constants $c, \sigma, C(\varepsilon_0) \in \mathbb{R}^+$ such that $C(\varepsilon_0) \leq c(2\varepsilon_0)^\sigma$, thanks to the assumption we made at first, and for every $\tau \in [0, T]$ it holds that

$$\|G(u_1(\tau)) - G(u_2(\tau))\| \leq C(\varepsilon_0) \|u_1(\tau) - u_2(\tau)\| \leq c(2\varepsilon_0)^\sigma \|u_1(\tau) - u_2(\tau)\|.$$

Hence we get

$$\|K(u_1(t)) - K(u_2(t))\| \leq c \int_0^t e^{-\theta(t-\tau)} (2\varepsilon_0)^\sigma \|u_1(\tau) - u_2(\tau)\| d\tau.$$

Like previously we use the inequality $\|u_1(\tau) - u_2(\tau)\| \leq e^{-\theta\tau} \|u_1 - u_2\|_{B_T}$ and we gain

$$\begin{aligned} \|K(u_1(t)) - K(u_2(t))\| &\leq \int_0^t e^{-\theta(t-\tau)} (2\varepsilon_0)^\sigma e^{-\theta\tau} \|u_1 - u_2\|_{B_T} d\tau \\ &\leq \int_0^t e^{-\theta t} (2\varepsilon_0)^\sigma \|u_1 - u_2\|_{B_T} d\tau \\ &\leq t(2\varepsilon_0)^\sigma \|u_1 - u_2\|_{B_T} e^{-\theta t}. \end{aligned}$$

Now by multiplying by $e^{\theta t}$ in both sides, we obtain

$$e^{\theta t} \|K(u_1(t)) - K(u_2(t))\| \leq t(2\varepsilon_0)^\sigma \|u_1 - u_2\|_{B_T}.$$

By taking the supremum on the interval $[0, T]$, we get

$$\|K(u_1) - K(u_2)\|_{B_T} \leq T(2\varepsilon_0)^\sigma \|u_1 - u_2\|_{B_T}.$$

Our aim is to prove that K is a contraction, so we have to impose simply that

$$T(2\varepsilon_0)^\sigma < 1$$

and, thanks to the choice of ε_0 , we can do it. Due to the Banach fixed-point Theorem 4.32, there exists a unique fixed point $u \in B_{B_T}(2\varepsilon_0)$ such that $K(u) = u$, that is

$$u(t) = K(u(t)) = e^{-At}u_0 + \int_0^t e^{-A(t-\tau)}G(u(\tau)) \, d\tau.$$

Thereby u is the solution of the Cauchy Problem (2.5) on the interval $[0, T]$.

Now we have to extend the solution. Indeed, by uniqueness, two solutions must agree on their common interval of definition. So if we have u_1 solution of the Cauchy Problem (2.5) on $[0, T_1]$ and u_2 solution of the Cauchy Problem (2.5) on $[0, T_2]$ with $0 < T_1 < T_2$, then, by uniqueness, $u_2|_{[0, T_1]}$ is a solution of (2.5) on $[0, T_1]$, therefore it has to coincide with u_1 . This reasoning can be made for every $T > 0$, hence the solution can be extended to $[0, +\infty)$, and the claim follows. □

Numerical simulations

The aim of this chapter is to apply a simplification of the model discussed in the previous chapters to the COVID-19 epidemic situation in Italy from 02/22/2020, a day when the epidemic began to spread, until the second wave on 10/31/2020, shortly before the introduction of the vaccination campaign.

3.1 Simplification of the model

The model used in the algorithm is as follows:

$$\begin{aligned}\frac{dS}{dt} &= -\lambda SI \\ \frac{dI}{dt} &= \lambda SI - \mu_{IR}I - \mu_{ID}I \\ \frac{dR}{dt} &= \mu_{IR}I \\ \frac{dD}{dt} &= \mu_{ID}I\end{aligned}$$

and we rewrite it by highlighting the population of Italy which we indicate with N

$$\begin{aligned}\frac{dS}{dt} &= -\frac{\beta}{N}SI \\ \frac{dI}{dt} &= \frac{\beta}{N}SI - \gamma I - \delta I \\ \frac{dR}{dt} &= \gamma I \\ \frac{dD}{dt} &= \delta I.\end{aligned}\tag{3.1}$$

Unlike what was done in the previous chapters, compartment C (carriers) is not present because the relative cumulative real data of the situation in Italy are not present (or we were unable to find it). The data on the situation in Italy were collected from the GitHub database https://github.com/CSSEGISandData/COVID-19/tree/master/csse_covid_19_data/csse_covid_19_time_series.

3.2 Presentation of the algorithm

At first, we explain the goal of the algorithm and then each line of the code and its functionalities. Our intention was to derive the parameters, that should have been part of the model capable of describing and representing the COVID-19 epidemic in Italy from 02/22/2020 to 10/31/2020, and to compare the optimal solutions obtained from our differential equation model with the real daily data from Italy. Next, we put the results obtained on graphs so as to visually display what was said. To do this, we first read the real and cumulative data of infected, recovered, and deceased people. Subsequently, by giving arbitrary parameters as input (therefore chosen randomly) we solved the system of differential equations thus obtaining solutions that we will then compare with the real ones.

```

1 from scipy.integrate import solve_ivp
2 from scipy.optimize import minimize
3 import numpy as np
4 import pandas as pd
5 import matplotlib.pyplot as plt

```

First of all we import some libraries, that is `solve_ivp` and `minimize` from `scipy` for solving differential equations and optimization, `numpy` for numerical operations, `pandas` for data manipulation, and `matplotlib.pyplot` for plotting.

```

1 # Load the datasets
2 confirmed=pd.read_csv("/Users/andreatedesco/Desktop/
   ↪ data/covid19_confirmed.csv")
3 deaths=pd.read_csv("/Users/andreatedesco/Desktop/
   ↪ data/covid19_deaths.csv")
4 recovered=pd.read_csv("/Users/andreatedesco/Desktop/
   ↪ data/covid19_recovered.csv")

```

Then, we read three CSV files containing COVID-19 data (confirmed cases, deaths, and recoveries) into pandas DataFrames.

```

1 # Define the country you're interested in (e.g., "Italy")
2 country = "Italy" # You can replace this with the name of the
   ↪ region or country you're interested in

```

We define the country of interest (Italy in this case).

```

1 # Find the row corresponding to the specific country
2 country_confirmed_row =
   ↪ confirmed.loc[confirmed['Country/Region'] ==
   ↪ country].iloc[0]
3 country_deaths_row = deaths.loc[deaths['Country/Region'] ==
   ↪ country].iloc[0]
4 country_recovered_row =
   ↪ recovered.loc[recovered['Country/Region'] ==
   ↪ country].iloc[0]

```

We extract the rows corresponding to Italy from each DataFrame.

```

1 # Define the range of columns you're interested in (from a start
   ↪ column to an end column)
2 start_column = '2/22/20' # The starting column (date)
3 end_column = '10/31/20' # The ending column (date)

```

We specify the date range of interest (from February 22, 2020, to October 31, 2020).

```

1 # Slice the DataFrame to get the columns within this range
2 selected_confirmed_columns =
   ↪ country_confirmed_row.loc[start_column:end_column].values.astype(float)
3 selected_deaths_columns =
   ↪ country_deaths_row.loc[start_column:end_column].values.astype(float)
4 selected_recovered_columns =
   ↪ country_recovered_row.loc[start_column:end_co

```

We slice the DataFrames to get the data for the specified date range and convert it to float arrays.

```

1 # Population of Italy (example value, replace with actual
   ↪ population)
2 population = 59_641_488
3
4 # Initial conditions
5 initial_infected = selected_confirmed_columns[0]
6 initial_recovered = selected_recovered_columns[0]

```

```

7 initial_deceased = selected_deaths_columns[0]
8 initial_susceptible = population - initial_infected -
  ↪ initial_recovered - initial_deceased
9
10 # Time data
11 t_data = np.arange(len(selected_confirmed_columns))

```

We set the population of Italy and calculate initial values for infected, recovered, deceased, and susceptible individuals based on the first data point in the selected date range. Then, we create an array of time points corresponding to the length of the selected date range.

```

1 # SIRD model
2 def sird_model(t, y, beta, gamma, delta):
3     S, I, R, D = y
4     dSdt = -beta * S * I / population
5     dIdt = beta * S * I / population - gamma * I - delta * I
6     dRdt = gamma * I
7     dDdt = delta * I
8     return [dSdt, dIdt, dRdt, dDdt]

```

The `sird_model` function defines the SIRD (Susceptible, Infected, Recovered, Deceased) model. It takes time t , state variables y (which include S , I , R , D), and parameters β , γ , and δ . The function returns the derivatives of S , I , R , and D with respect to time.

```

1 # Compute the number of susceptibles every day
2 susceptibles_real = population - selected_confirmed_columns -
  ↪ selected_recovered_columns - selected_deaths_columns
3
4 # Calculate the number of new infected people every day and
  ↪ ensure no negative values
5 new_infected_real = np.diff(selected_confirmed_columns,
  ↪ prepend=selected_confirmed_columns[0])
6 new_infected_real = np.maximum(new_infected_real, 0) # Replace
  ↪ negative values with 0
7
8 # Calculate the number of new recovered people every day and
  ↪ ensure no negative values
9 new_recovered_real = np.diff(selected_recovered_columns,
  ↪ prepend=selected_recovered_columns[0])
10 new_recovered_real = np.maximum(new_recovered_real, 0) #
  ↪ Replace negative values with 0

```

```

11  # Calculate the number of new deceased people every day and
    ↪ ensure no negative values
12  new_deceased_real = np.diff(selected_deaths_columns,
    ↪ prepend=selected_deaths_columns[0])
13  new_deceased_real = np.maximum(new_deceased_real, 0)  # Replace
    ↪ negative values with 0

```

We calculate the number of susceptible individuals each day by subtracting the confirmed, recovered, and deceased cases from the total population. We compute the daily new infected cases using the difference between consecutive days' confirmed cases and we ensure no negative values by replacing them with 0. We compute the daily new recovered cases similarly, ensuring no negative values. Finally, we compute the daily new deceased cases, also ensuring no negative values.

```

1  # Mean Squared Logarithmic Error (MSLE) function for daily data
2  def msle_daily(I_pred, I_data, R_pred, R_data, D_pred, D_data):
3      # Calculate daily changes for model predictions
4      new_I_pred = np.diff(I_pred, prepend=I_pred[0])
5      new_I_pred = np.maximum(new_I_pred, 0)  # Replace negative
        ↪ values with 0
6
7      new_R_pred = np.diff(R_pred, prepend=R_pred[0])
8      new_R_pred = np.maximum(new_R_pred, 0)  # Replace negative
        ↪ values with 0
9
10     new_D_pred = np.diff(D_pred, prepend=D_pred[0])
11     new_D_pred = np.maximum(new_D_pred, 0)  # Replace negative
        ↪ values with 0
12
13     # Calculate daily changes for real data
14     new_I_data = np.diff(I_data, prepend=I_data[0])
15     new_I_data = np.maximum(new_I_data, 0)  # Replace negative
        ↪ values with 0
16
17     new_R_data = np.diff(R_data, prepend=R_data[0])
18     new_R_data = np.maximum(new_R_data, 0)  # Replace negative
        ↪ values with 0
19
20     new_D_data = np.diff(D_data, prepend=D_data[0])
21     new_D_data = np.maximum(new_D_data, 0)  # Replace negative
        ↪ values with 0
22

```

```

23     # Compute MSLE error for daily data
24     error = np.mean((np.log(new_I_pred + 1) - np.log(new_I_data
    ↪ + 1)) ** 2) + \
25         np.mean((np.log(new_R_pred + 1) - np.log(new_R_data
    ↪ + 1)) ** 2) + \
26         np.mean((np.log(new_D_pred + 1) - np.log(new_D_data
    ↪ + 1)) ** 2)
27
28     return error

```

This part of the code defines a function to calculate the Mean Squared Logarithmic Error (MSLE) for daily data. The `msle_daily` function takes in predicted and actual data for infected (`I_pred`, `I_data`), recovered (`R_pred`, `R_data`), and deceased (`D_pred`, `D_data`) individuals. Then it computes the daily changes in the predicted data using `np.diff`, which calculates the difference between consecutive elements. It ensures no negative values by replacing them with 0 using `np.maximum`. Similarly, it computes the daily changes in the actual data and ensures no negative values. It calculates the MSLE for the daily changes in infected, recovered, and deceased individuals. The MSLE is computed by taking the mean of the squared differences between the logarithms of the predicted and actual values (plus 1 to avoid $\log(0)$). Eventually, it returns the computed MSLE error. This function helps evaluate the accuracy of your model by comparing the predicted daily changes to the actual daily changes in the data.

```

1  # Objective function with MSLE to improve fit using daily data
2  def objective(params, t_data, I_data, R_data, D_data):
3      beta, gamma, delta = params
4      initial_conditions = [initial_susceptible, initial_infected,
    ↪ initial_recovered, initial_deceased]
5
6      # Solve the SIRD model
7      sol = solve_ivp(sird_model, [t_data[0], t_data[-1]],
    ↪ initial_conditions, args=(beta, gamma, delta),
    ↪ t_eval=t_data)
8
9      # Extract predicted values
10     I_pred, R_pred, D_pred = sol.y[1], sol.y[2], sol.y[3]
11
12     # Convert data to float
13     I_data = I_data.astype(float)
14     R_data = R_data.astype(float)
15     D_data = D_data.astype(float)

```

```

16
17     # Compute MSLE error using daily data
18     error = msle_daily(I_pred, I_data, R_pred, R_data, D_pred,
19                        ↪ D_data)
19
20     return error

```

This part of the code defines an objective function to optimize the parameters of the SIRD model using the Mean Squared Logarithmic Error (MSLE) for daily data. The `objective` function takes in parameters (`params`), time data (`t_data`), and actual data for infected (`I_data`), recovered (`R_data`), and deceased (`D_data`) individuals. We extract the parameters β , γ , and δ from the `params` array. Then, we set the initial conditions for the SIRD model using the previously defined initial values for susceptible, infected, recovered, and deceased individuals. We use `solve_ivp` to solve the SIRD model over the time range specified by `t_data`. The solution `sol` contains the predicted values for susceptible, infected, recovered, and deceased individuals over time. We extract the predicted values for infected (`I_pred`), recovered (`R_pred`), and deceased (`D_pred`) individuals from the solution. We ensure the actual data arrays are of type float for accurate calculations. We call the `msle_daily` function to compute the MSLE error between the predicted and actual daily changes in infected, recovered, and deceased individuals. Eventually, the code returns the computed MSLE error, which will be used to optimize the parameters. This function is crucial for fitting the SIRD model to the actual data by minimizing the MSLE error.

```

1  # Initial guess for parameters
2  initial_guess = [0.45, 0.1, 0.01]

```

We set an initial guess for the parameters β , γ , and δ . These parameters are completely arbitrary. We have tested several of them without obtaining very different results. That is the reason why we left them.

```

1  # Optimize parameters with more iterations and tighter
   ↪ tolerances for better results
2  result = minimize(objective, initial_guess,
3                    args=(t_data, selected_confirmed_columns,
4                          ↪ selected_recovered_columns,
5                          ↪ selected_deaths_columns),
6                    method='L-BFGS-B',
5                    bounds=[(0, 1), (0, 1), (0, 1)],
6                    options={'maxiter': 10000, 'maxfun': 20000})

```

We use the `minimize` function from `scipy.optimize` to optimize the parameters. The objective function is minimized using the 'L-BFGS-B' method, which is suitable for bound-constrained optimization. The bounds for the parameters are set between 0 and 1. The options dictionary specifies a high number of iterations (`maxiter`) and function evaluations (`maxfun`) to ensure a thorough optimization process.

```

1 # Solve the model with optimized parameters
2 beta_opt, gamma_opt, delta_opt = result.x
3 sol = solve_ivp(sird_model, [t_data[0], t_data[-1]],
  ↪ [initial_susceptible, initial_infected, initial_recovered,
  ↪ initial_deceased], args=(beta_opt, gamma_opt, delta_opt),
  ↪ t_eval=t_data)

```

Now we extract the optimized parameters `beta_opt`, `gamma_opt`, and `delta_opt` from the optimization result. We solve the SIRD model again using `solve_ivp` with the optimized parameters over the specified time range. This part of the code fine-tunes the model to better fit the actual data by optimizing the parameters, and then it solves the model with these optimized parameters to predict the epidemic's progression.

```

1 # Extract predicted values
2 S_pred, I_pred, R_pred, D_pred = sol.y
3
4 daily_infected_pred = np.diff(I_pred, prepend=I_pred[0])
5 daily_infected_pred = np.maximum(daily_infected_pred, 0) #
  ↪ Replace negative values with 0
6
7 daily_recovered_pred = np.diff(R_pred, prepend=R_pred[0])
8 daily_recovered_pred = np.maximum(daily_recovered_pred, 0) #
  ↪ Replace negative values with 0
9
10 daily_deceased_pred = np.diff(D_pred, prepend=D_pred[0])
11 daily_deceased_pred = np.maximum(daily_deceased_pred, 0) #
  ↪ Replace negative values with 0
12

```

This part of the code calculates the daily changes from the model predictions.

```

1 plt.figure(figsize=(12, 8))
2 plt.plot(t_data, new_infected_real, label='New Infected People',
  ↪ color='purple', linewidth=2)
3 plt.xlabel('Days', fontsize=14) # Adjust the font size for the
  ↪ x-axis label

```

```

4 plt.ylabel('Number of New Infected People', fontsize=18) #
  ↳ Adjust the font size for the y-axis label
5 plt.title('Number of New Infected People Every Day',
  ↳ fontsize=18) # Adjust the font size for the title
6 plt.xticks(fontsize=16) # Adjust the font size for the x-axis
  ↳ tick labels
7 plt.yticks(fontsize=16) # Adjust the font size for the y-axis
  ↳ tick labels
8 plt.legend()
9 plt.grid(True)
10 plt.show()
11
12 plt.figure(figsize=(12, 8))
13 plt.plot(t_data, new_recovered_real, label='New Recovered
  ↳ People', color='purple', linewidth=2)
14 plt.xlabel('Days', fontsize=14) # Adjust the font size for the
  ↳ x-axis label
15 plt.ylabel('Number of New Recovered People', fontsize=18) #
  ↳ Adjust the font size for the y-axis label
16 plt.title('Number of New Recovered People Every Day',
  ↳ fontsize=18) # Adjust the font size for the title
17 plt.xticks(fontsize=16) # Adjust the font size for the x-axis
  ↳ tick labels
18 plt.yticks(fontsize=16) # Adjust the font size for the y-axis
  ↳ tick labels
19 plt.legend()
20 plt.grid(True)
21 plt.show()
22
23 plt.figure(figsize=(12, 8))
24 plt.plot(t_data, new_deceased_real, label='New Deceased People',
  ↳ color='purple', linewidth=2)
25 plt.xlabel('Days', fontsize=14) # Adjust the font size for the
  ↳ x-axis label
26 plt.ylabel('Number of New Deceased People', fontsize=18) #
  ↳ Adjust the font size for the y-axis label
27 plt.title('Number of New Deceased People Every Day',
  ↳ fontsize=18) # Adjust the font size for the title
28 plt.xticks(fontsize=16) # Adjust the font size for the x-axis
  ↳ tick labels
29 plt.yticks(fontsize=16) # Adjust the font size for the y-axis
  ↳ tick labels
30 plt.legend()

```

```

31 plt.grid(True)
32 plt.show()

```

We create figures with a specified size. The first one plots the number of new infected people each day. We add labels for the x and y axes, a title, a legend, and a grid for better readability. Then we display the plot. We do the same for the number of recovered and deceased people each day.

```

1  # Plot error over time
2  error_over_time_pred = (np.log(daily_infected_pred + 1) -
   ↪ np.log(new_infected_real + 1)) ** 2 + \
3                               (np.log(daily_recovered_pred + 1) -
   ↪ np.log(new_recovered_real + 1)) ** 2 + \
4                               (np.log(daily_deceased_pred + 1) -
   ↪ np.log(new_deceased_real + 1)) ** 2
5
6  plt.figure(figsize=(12, 8))
7  plt.plot(t_data, error_over_time_pred, label='Error Over Time',
   ↪ color='orange', linewidth=2)
8  plt.xlabel('Days', fontsize=16) # Adjust the font size for the
   ↪ x-axis label
9  plt.ylabel('Error', fontsize=16) # Adjust the font size for the
   ↪ y-axis label
10 plt.title('Error Over Time', fontsize=18) # Adjust the font
   ↪ size for the title
11 plt.xticks(fontsize=14) # Adjust the font size for the x-axis
   ↪ tick labels
12 plt.yticks(fontsize=14) # Adjust the font size for the y-axis
   ↪ tick labels
13 plt.legend()
14 plt.grid(True)
15 plt.show()

```

This plot helps to see how the model's predictions compare to the actual data over the specified period, highlighting any discrepancies. A high error at certain points suggests periods where the model performs poorly, which could indicate specific times where assumptions or parameters may need refinement. A consistent error reduction over time might suggest that the model captures the dynamics better in later stages.

```

1  # Plotting the comparison between real data and model
   ↪ predictions for daily infected cases
2  plt.figure(figsize=(10, 6))

```

```

3 plt.plot(t_data[zoom_start:zoom_end],
  ↪ daily_infected_pred[zoom_start:zoom_end], label='Daily
  ↪ Infected (Pred)', color='blue')
4 plt.plot(t_data[zoom_start:zoom_end],
  ↪ new_infected_real[zoom_start:zoom_end], label='Daily
  ↪ Infected Italy', color='purple')
5 plt.xlabel('Days since start', fontsize=16) # Adjust the font
  ↪ size for the x-axis label
6 plt.ylabel('Number of Daily Infected People', fontsize=16) #
  ↪ Adjust the font size for the y-axis label
7 plt.title('Comparison of Daily Infected People: Pred vs Italy',
  ↪ fontsize=18) # Adjust the font size for the title
8 plt.xticks(fontsize=16) # Adjust the font size for the x-axis
  ↪ tick labels
9 plt.yticks(fontsize=16) # Adjust the font size for the y-axis
  ↪ tick labels
10 plt.legend()
11 plt.show()

1 # Plotting the comparison between real data and model
  ↪ predictions for daily recovered cases
2 plt.figure(figsize=(10, 6))
3 plt.plot(t_data[zoom_start:zoom_end],
  ↪ daily_recovered_pred[zoom_start:zoom_end], label='Daily
  ↪ Recovered (Pred)', color='blue')
4 plt.plot(t_data[zoom_start:zoom_end],
  ↪ new_recovered_real[zoom_start:zoom_end], label='Daily
  ↪ Recovered Italy', color='purple')
5 plt.xlabel('Days since start', fontsize=16) # Adjust the font
  ↪ size for the x-axis label
6 plt.ylabel('Number of Daily Recovered People', fontsize=16) #
  ↪ Adjust the font size for the y-axis label
7 plt.title('Comparison of Daily Recovered People: Pred vs Italy',
  ↪ fontsize=18) # Adjust the font size for the title
8 plt.xticks(fontsize=16) # Adjust the font size for the x-axis
  ↪ tick labels
9 plt.yticks(fontsize=16) # Adjust the font size for the y-axis
  ↪ tick labels
10 plt.legend()
11 plt.show()

1 # Plotting the comparison between real data and model
  ↪ predictions for daily deceased cases

```

```
2 plt.figure(figsize=(10, 6))
3 plt.plot(t_data[zoom_start:zoom_end],
    ↪ daily_deceased_pred[zoom_start:zoom_end], label='Daily
    ↪ Deceased (Pred)', color='blue')
4 plt.plot(t_data[zoom_start:zoom_end],
    ↪ new_deceased_real[zoom_start:zoom_end], label='Daily
    ↪ Deceased Italy', color='purple')
5 plt.xlabel('Days since start', fontsize=16) # Adjust the font
    ↪ size for the x-axis label
6 plt.ylabel('Number of Daily Deceased People', fontsize=16) #
    ↪ Adjust the font size for the y-axis label
7 plt.title('Comparison of Daily Deceased People: Pred vs Italy',
    ↪ fontsize=18) # Adjust the font size for the title
8 plt.xticks(fontsize=16) # Adjust the font size for the x-axis
    ↪ tick labels
9 plt.yticks(fontsize=16) # Adjust the font size for the y-axis
    ↪ tick labels
10 plt.legend()
11 plt.show()
```

This code snippet generates comparative visualizations for the mathematical model's predictions versus real-world data from Italy, focusing on three key metrics: daily infected, daily recovered, and daily deceased cases. By visually comparing the predicted and real values, we can identify discrepancies, validate the model's assumptions, and potentially refine its parameters for better accuracy.

3.3 Results

In this section, we will present the results we got and we will comment and describe the graphs we obtained.

The outcomes presented in this section highlight the comparison between the predictions of the proposed mathematical model and the actual epidemiological data from Italy. Specifically, the daily numbers of infected, recovered, and deceased individuals are analyzed to evaluate the model's accuracy and reliability. Graphical representations of these comparisons reveal the extent to which the model captures the temporal dynamics of the epidemic, showcasing its strengths in replicating observed trends and identifying areas where discrepancies occur. Additionally, the evolution of the error over time is examined to provide a quantitative assessment of the model's predictive performance, offering insights into periods of alignment and divergence. These analyses serve as the foundation for drawing meaningful conclusions about the model's applicability and potential refinements.

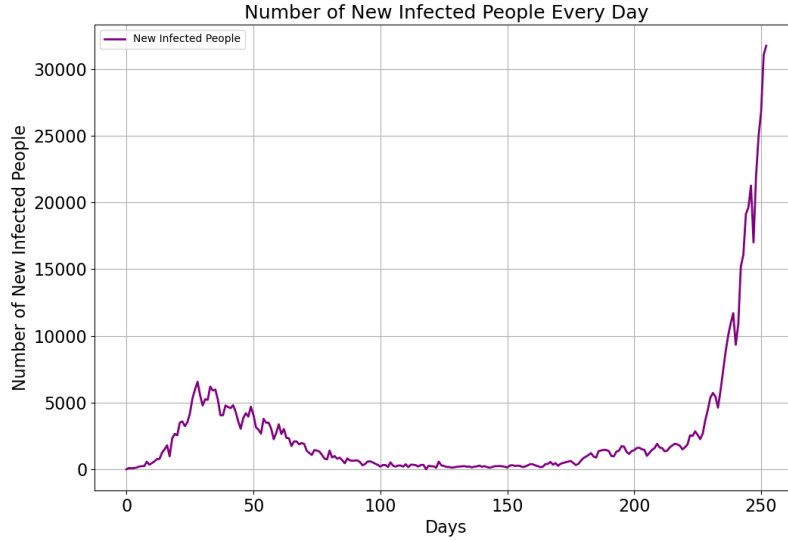


Figure 3.1: Daily infected in Italy

We can notice the presence of two waves, the first one immediately at the beginning of the epidemic and the second one which occurs after the end of the summer.

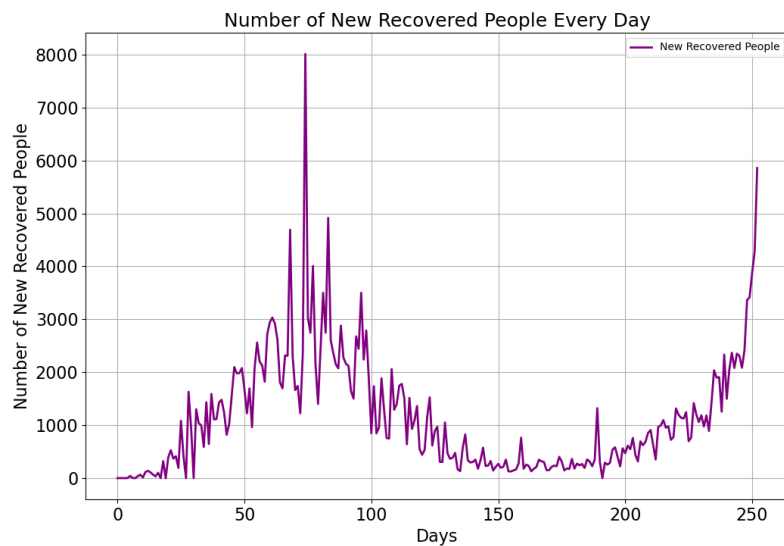


Figure 3.2: Daily recovered in Italy

The number of recoveries increased drastically initially due to the restrictive lockdown measures and subsequently decreased as a consequence of the decrease in infected people.

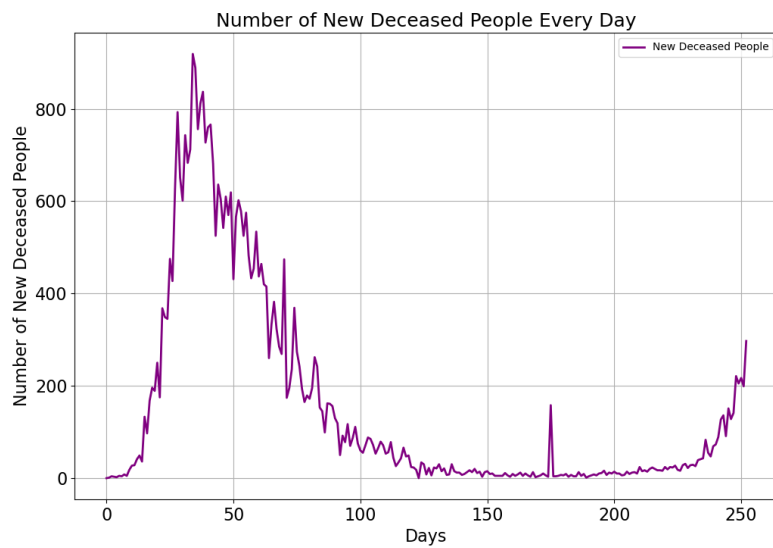


Figure 3.3: Daily deceased in Italy

Even with regards to deaths, there was a first peak during the first wave, in conjunction with the first restrictive measures such as the lockdown, and a second peak on the way.

These are the data on the real situation in Italy during the COVID-19 epidemic. Now we will compare these data with those of the optimal model which should therefore best simulate the graphs just shown.

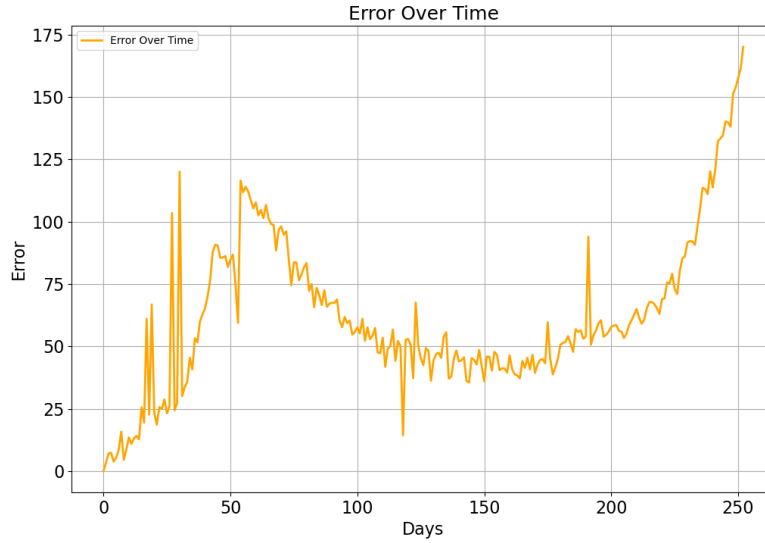


Figure 3.4: Error Over Time

We can see that over time the error increases and then decreases again and finally grows again. This graph allows us to immediately understand the difficulties of the model in faithfully simulating the current situation in Italy. We have associated this explanation with the elevated simplification of the model used, caused by the lack of data and the complexity of the situation. We now report the graphs where we compare the real data of COVID-19 in Italy and those obtained as a solution of the model with the optimal parameters, which are

$$\beta = 3.56159716 \times 10^{-1}, \quad \gamma = 5.97309077 \times 10^{-2}, \quad \delta = 7.25073045 \times 10^{-5}.$$

We will compare the data obtained only for a very short period of time, namely the first 20 days after the outbreak of the epidemic. In fact, due to the numerous simplifications, the model is not able to slavishly simulate the course of the disease. A long-term comparison would have highlighted this discrepancy disproportionately.

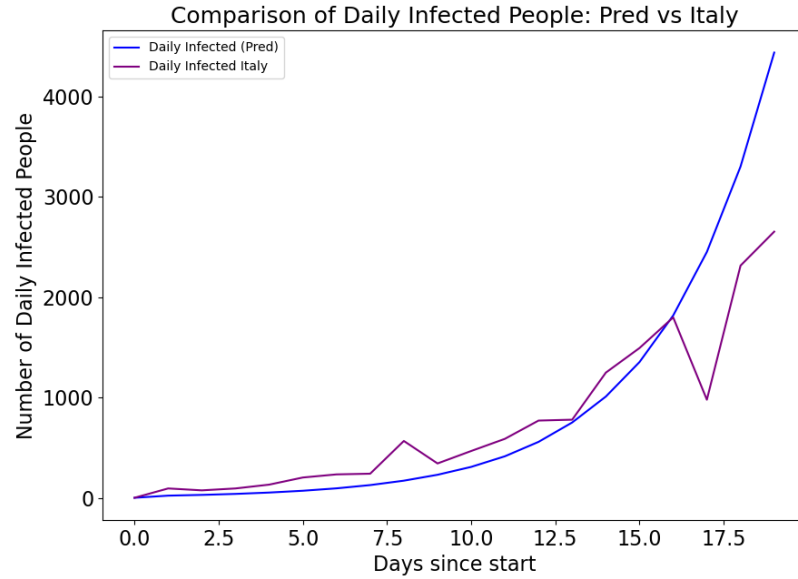


Figure 3.5: Comparison of Daily Infected People: Pred vs Italy

From the previous graph, we can see the powerful growth of the infected in the predicted model, even if it seems that the simulation of this compartment is happening as hoped.

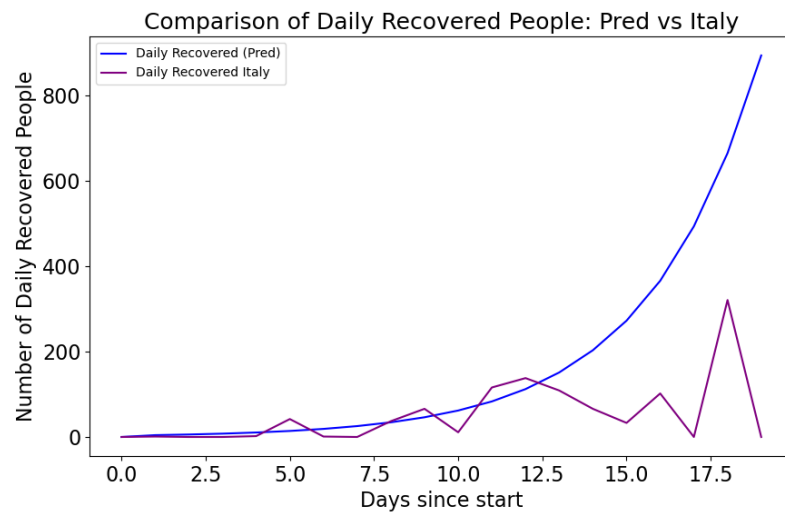


Figure 3.6: Comparison of Daily Recovered People: Pred vs Italy

From this graph, however, we can already begin to observe the distance between the predicted data and the real ones. A comparison over a longer period of time would have highlighted this difference even more.

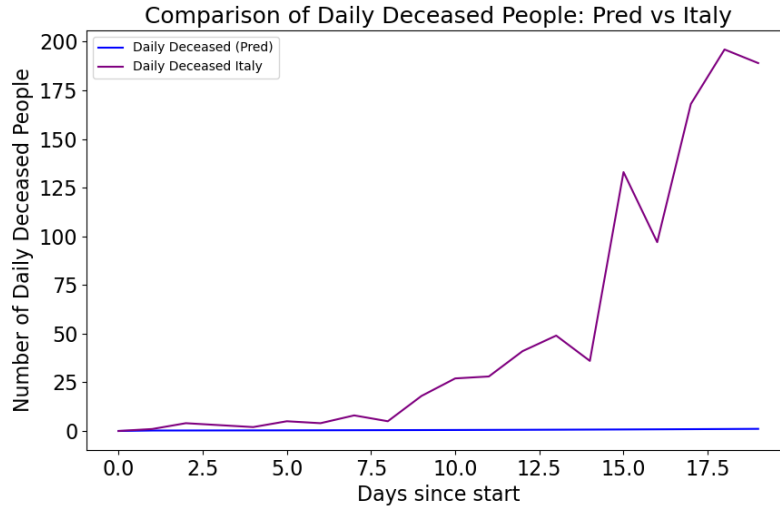


Figure 3.7: Comparison of Daily Deceased People: Pred vs Italy

From the latter, we note an absolute difference. In the predicted model the number of daily deaths is close to zero in the first 20 days of the epidemic. This trend is probably due to the optimal death rate that was calculated by the algorithm. This last graph allows us to understand how far, in reality, the model used is from the one that actually describes the spread of the disease in Italy.

3.4 Conclusions

The analysis of the real COVID-19 data in Italy and its comparison with the predictions of the optimal SIRD model reveal notable discrepancies that emphasize the limitations of the current framework. The algorithm captures broad trends, such as the presence of two epidemic waves—the first occurring at the onset of the epidemic and the second following the summer period—and the resulting peaks in recoveries and deaths. However, significant deviations emerge upon closer inspection. Specifically, the model struggles to accurately represent the mortality trends during the first 20 days, where the predicted number of daily deaths remains close to zero. This discrepancy is likely a

consequence of the optimal death rate calculated by the parameter estimation algorithm, which fails to fully account for the real dynamics during the early stages of the epidemic. Similarly, while the predicted growth of infections mirrors the observed general trend, the model exhibits increasing error over time, which later decreases before rising again. This pattern highlights the difficulty in simulating the complex interplay of factors influencing disease spread, such as changing public health policies, population behavior, and data inaccuracies.

These discrepancies stem primarily from the elevated simplifications of the model, necessitated by limited data availability and the inherent complexity of epidemic dynamics. For example, the exclusion of external factors such as mobility patterns, age-dependent susceptibility, and spatial heterogeneity restricts the model's accuracy. Additionally, the absence of compartment for hospitalizations further limits the scope of the analysis, particularly in the later phases of the epidemic. A longer comparison period between real and simulated data would likely accentuate these discrepancies, underscoring the need for further model refinements.

Despite these limitations, this work lays a strong foundation for future improvements and research. Several promising avenues exist to enhance the model's predictive power and versatility. First, incorporating additional compartments, such as vaccinated, hospitalized, and reinfected individuals, would allow the model to better reflect real-world disease dynamics. Second, introducing time-dependent parameters that adapt to evolving public health measures, such as lockdowns, vaccination campaigns, and behavioral changes, could significantly improve the model's realism. Moreover, a better optimization process could be implemented by iterating on the current parameter estimation approach, refining the results to minimize residual error more effectively. From a computational perspective, the use of machine learning techniques could enhance parameter estimation, reduce error margins, and improve predictive accuracy over longer time horizons. Furthermore, exploring stochastic variations of the SIRCD model would help account for the role of randomness in disease spread, offering a more nuanced understanding of epidemic dynamics.

The theoretical analysis of the SIRCD model reveals significant insights into its dynamic behavior. The proof of the global existence of solutions establishes that the system is well-posed, ensuring that its predictions remain reliable over time without divergence. Stability analysis of equilibrium points further highlights the conditions under which the epidemic transitions toward disease-free or endemic states. These findings provide a robust foundation for deeper exploration through bifurcation analysis, which investigates how variations in key parameters, such as the basic reproduction number (R_0) or contact rates, induce qualitative changes in the system's dynamics. For instance, the emergence of backward bifurcation, where stable disease-free and endemic

equilibria coexist, underscores the complexity of controlling an epidemic. Such analyses complement the stability results and enrich our understanding of the model's sensitivity to external factors, guiding theoretical development and practical application.

From a geographical standpoint, coupling the model with real-time spatial data would enable the study of disease propagation across regions, providing valuable insights into local transmission patterns and the effectiveness of regional interventions. Additionally, the model's adaptability can be tested by applying it to other infectious diseases, such as seasonal influenza or emerging epidemics, to evaluate its performance in several epidemiological contexts. Finally, future research could explore the effects of waning immunity and reinfections, factors that play a crucial role in the long-term evolution of epidemics.

In conclusion, while the current model demonstrates clear limitations in faithfully simulating the real data, it provides a robust starting point for deeper investigations. By addressing these discrepancies and embracing more sophisticated methods, this framework has the potential to become a powerful tool for understanding and managing infectious diseases, contributing meaningfully to public health planning and preparedness for future outbreaks.

Appendix A

4.1 Theory of Ordinary Differential Equations

Let $F : \Omega \subseteq \mathbb{R}^{n+2} \rightarrow \mathbb{R}$, with $\Omega \neq \emptyset$ being an open and connected set, and $n \geq 1$ an integer. We define an ordinary differential equation of order n as a relation of the form:

$$F\left(x, u(x), u'(x), \dots, u^{(n)}(x)\right) = 0, \quad (4.1)$$

where $u^{(j)}(x)$, for $j = 1, \dots, n$, represents the j -th derivative of the function $u(x)$.

Definition 4.1. Let I be an open interval in $\mathbb{R}$. We define a solution or integral of the ordinary differential equation as a function $u = u(x)$ such that:

$$u(x) \in C^n(I), \quad F\left(x, u(x), u'(x), \dots, u^{(n)}(x)\right) = 0 \quad \forall x \in I$$

An ordinary differential equation is said to be **autonomous** if F does not explicitly depend on x , that is, equation (4.1) becomes:

$$F\left(u(x), u'(x), \dots, u^{(n)}(x)\right) = 0, \quad (4.2)$$

The ordinary differential equation (4.1) is said to be in **normal form** if it can be written explicitly in terms of $u^{(n)}(x)$:

$$u^{(n)}(x) = G\left(x, u, u', \dots, u^{(n-1)}\right). \quad (4.3)$$

An ordinary differential equation is called **linear** if F is a linear combination of $u, u', \dots, u^{(n)}$, that is:

$$F(x, u, u', \dots, u^{(n)}) = s(x) + b_0(x)u + b_1(x)u' + \dots + b_n(x)u^{(n)}$$

or, equivalently, the equation (4.1) can be written as:

$$u^{(n)} = \sum_{i=0}^{n-1} a_i(x)u^{(i)} + f(x)$$

where: $f(x), a_0(x), a_1(x), \dots, a_{n-1}(x) \in C^0(I)$.

The term $f(x)$ is called the **forcing term**, and if it is zero, the linear differential equation is said to be **homogeneous**.

Let $f : \Omega \subseteq \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n$, with $\Omega \neq \emptyset$ an open and connected set, and $n \geq 1$ an integer. We define a system of first-order ordinary differential equations in normal form as follows:

$$u' = f(x, u) \tag{4.4}$$

Let I be an open interval in $\mathbb{R}$ and $\omega \subseteq \mathbb{R}^n$, with $\omega \neq \emptyset$, an open and connected set such that:

$$I \times \omega \subseteq \Omega.$$

We define a solution or integral of the system of equations (4.4) as a function:

$$u = u(x) : I \rightarrow \omega$$

such that:

$$u(x) \in C^m(I; \mathbb{R}^n), \quad u'(x) = f(x, u(x)) \quad \forall x \in I.$$

Every ordinary differential equation of order n can be reduced to a system of first-order ODEs. This is accomplished by introducing new variables that represent the successive derivatives of the unknown function $u(x)$.

Consider the ordinary differential equation of order n :

$$u^{(n)}(x) = G(x, u(x), u'(x), \dots, u^{(n-1)}(x)).$$

We can define a new system of variables as follows:

$$y_1 = u(x), \quad y_2 = u'(x), \quad \dots, \quad y_n = u^{(n-1)}(x).$$

By differentiating each new variable, we obtain a system of first-order differential equations:

$$y'_1 = y_2, \quad y'_2 = y_3, \quad \dots, \quad y'_{n-1} = y_n,$$

and

$$y'_n = G(x, y_1, y_2, \dots, y_n).$$

Thus, the higher-order differential equation has been transformed into a system of first-order equations, which can be analyzed using the methods applicable to first-order systems.

Theorem 4.2 (Picard–Lindelöf Theorem). *Let f be a function defined on a neighborhood Ω of $(t_0, u_0) \in \mathbb{R} \times \mathbb{R}^n$ where Ω is*

$$\Omega = I \times J = \{(t, u) \in \mathbb{R} \times \mathbb{R}^n : |t - t_0| \leq a, \|u - u_0\| \leq b\}$$

with a, b positive real numbers. Suppose $f \in C^0(\Omega)$. Assume also that f is a Lipschitz function with respect to u and uniformly continuous with respect to t :

$$\|f(t, u) - f(t, v)\| \leq L\|u - v\| \quad \forall t \in I, \forall u, v \in J$$

with $L > 0$ Lipschitz constant. Under these conditions, there exists $T > 0$, such that the Cauchy Problem

$$\begin{cases} u'(t) = f(t, u(t)), & |t - t_0| < T; \\ u(t_0) = u_0 \end{cases}$$

has a unique solution $u(t) \in C^1((t_0 - T, t_0 + T; J))$.

Proof. In [4, Section 1.4] there is a proof of this Theorem. □

Now, suppose that Ω is an open subset of $\mathbb{R} \times \mathbb{R}^n$, $(t_0, a) \in \Omega$, and $f : \Omega \rightarrow \mathbb{R}^n$ is locally Lipschitz continuous with respect to x on Ω . Then the Picard–Lindelöf Theorem indicates that the Cauchy Problem

$$\begin{cases} \dot{x} = f(t, x) \\ x(t_0) = a \end{cases} \tag{4.5}$$

has a solution existing on some time interval containing t_0 in its interior and the solution is unique on that interval. Let's say that an interval of existence is an interval containing t_0 on which a solution of (4.5) exists. The following theorem indicates how large an interval of existence may be.

Theorem 4.3 (Maximal Interval of Existence). *The Cauchy Problem 4.5 has a maximal interval of existence, and it is of the form (ω, α) , with $\omega \in [-\infty, +\infty)$ and $\alpha \in (-\infty, +\infty]$. There is a unique solution $x(t)$ of 4.5 on (ω, α) , and $(t, x(t))$ leaves every compact subset K of Ω as $t \rightarrow \omega^+$ and as $t \rightarrow \alpha^-$.*

Proof. The source [4, Section 1.5] contains a proof of this theorem. □

Corollary 4.4. *If Ω_0 is a bounded set and $\Omega = (c, d) \times \Omega_0$ (with $c \in [-\infty, +\infty)$ and $d \in (-\infty, +\infty]$), then either $\alpha = d$ or $x(t) \rightarrow \partial\Omega_0$ as $t \rightarrow \alpha^-$, and either $\omega = c$ or $x(t) \rightarrow \partial\Omega_0$ as $t \rightarrow \omega^+$.*

Corollary 4.5. *If $\Omega = (c, d) \times \mathbb{R}^n$ with $(c \in [-\infty, +\infty)$ and $d \in (-\infty, +\infty])$, then either $\alpha = d$ or $|x(t)| \rightarrow +\infty$ as $t \rightarrow \alpha^-$, and either $\omega = c$ or $|x(t)| \rightarrow +\infty$ as $t \rightarrow \omega^+$.*

If we are dealing with an autonomous equation on a bounded set, then the first corollary applies to tell us that the only way a solution could fail to exist for all time is for it to approach the boundary of the spatial domain. (Note that this is not the same as saying that $x(t)$ converges to a particular point on the boundary;) The second corollary says that autonomous equations on all of $\mathbb{R}^n$ have solutions that exist until they become unbounded.

The comparison principle is a powerful tool in the qualitative analysis of differential equations, providing a method to compare solutions of different equations.

Lemma 4.6 (Comparison Principle I). *Let $y(t), Y(t) : [t_0, t_1] \rightarrow V \subseteq \mathbb{R}$ be differentiable functions, and let $f, F : [t_0, t_1] \times V \rightarrow \mathbb{R}$ be continuous functions such that*

$$f(t, y) \leq F(t, y), \quad \forall (t, y) \in [t_0, t_1] \times V$$

and suppose that:

$$y'(t) = f(t, y(t)), \quad \forall t \in [t_0, t_1],$$

$$Y'(t) = F(t, Y(t)), \quad \forall t \in [t_0, t_1].$$

If

$$y(t_0) \leq Y(t_0),$$

then

$$y(t) \leq Y(t), \quad \forall t \in [t_0, t_1].$$

This result ensures that if one function starts smaller than another and its growth is constrained in a specific way, the initial inequality is preserved over

time.

We also have the following variant of the comparison principle.

Lemma 4.7 (Comparison Principle II). *Let $y(t), Y(t) : [t_0, t_1] \rightarrow V \subseteq \mathbb{R}$ be differentiable functions, and let $f : [t_0, t_1] \times V \rightarrow \mathbb{R}$ be a continuous function such that:*

$$\begin{aligned} y'(t) &\leq f(t, y(t)), \quad \forall t \in [t_0, t_1], \\ Y'(t) &= f(t, Y(t)), \quad \forall t \in [t_0, t_1]. \end{aligned}$$

If

$$y(t_0) \leq Y(t_0),$$

then

$$y(t) \leq Y(t), \quad \forall t \in [t_0, t_1].$$

Two immediate consequences are the following:

Lemma 4.8 (Positivity Principle). *Let $y(t)$ be a solution to the differential equation*

$$y'(t) = f(t, y)$$

defined on an interval $[0, T)$, where $f(t, y)$ is continuous and satisfies the following conditions:

1. **Initial Condition:** *Let $y(0) = y_0 \geq 0$.*
2. **Non-negativity of the Function:** *Suppose $f(t, y)$ is continuous and satisfies $f(t, y) \geq 0$ for all t in $[0, T)$ and for all $y \geq 0$.*

Then, under these conditions, the solution $y(t)$ remains non-negative for all t in $[0, T)$:

$$y(t) \geq 0 \quad \text{for all } t \in [0, T).$$

Lemma 4.9 (Gronwall's Lemma). *Let I be a closed interval of the form $[a, \infty)$, $[a, b]$, or $[a, b)$ with $a < b$. Let $f(t), u(t) : I \rightarrow \mathbb{R}$ be continuous functions. If u is differentiable on the interior of I and satisfies*

$$u'(t) \leq f(t)u(t), \quad t \in \overset{\circ}{I},$$

then

$$u(t) \leq u(a) \exp \left(\int_a^t f(s) ds \right),$$

for all $t \in I$.

Consider the differential equation $\dot{x} = f(x)$ and its associated dynamical system $\varphi(t, x)$ on a phase space Ω .

Definition 4.10. We say that a point $x \in \Omega$ is an equilibrium point or a singular point or a stationary point if $f(x) = 0$. For such a point, $\varphi(t, x) = x$ for all $t \in \mathbb{R}$.

Definition 4.11. A point $x \in \Omega$ that is not a singular point is called a regular point.

One important aspect of the behavior of flows has to do with whether solutions that start near a given solution stay near it for all time and/or move closer to it as time elapses. This question, which is the subject of stability theory, is not just of interest when the given solution corresponds to an equilibrium solution, so we study it—initially, at least—in a fairly broad context.

First, we define some types of stability for solutions of the (possibly) nonautonomous equation

$$\dot{x} = f(t, x). \quad (4.6)$$

Definition 4.12. A solution $\bar{x}(t)$ of (4.6) is (Lyapunov) stable if for each $\varepsilon > 0$ and $t_0 \in \mathbb{R}$ there exists $\delta = \delta(\varepsilon, t_0) > 0$ such that if $x(t)$ is a solution of (4.6) and $|x(t_0) - \bar{x}(t_0)| < \delta$ then $|x(t) - \bar{x}(t)| < \varepsilon$ for all $t \geq t_0$.

Definition 4.13. A solution $\bar{x}(t)$ of (4.6) is asymptotically stable if it is (Lyapunov) stable and if for every $t_0 \in \mathbb{R}$ there exists $\delta = \delta(t_0) > 0$ such that if $x(t)$ is a solution of (4.6) and $|x(t_0) - \bar{x}(t_0)| < \delta$ then $|x(t) - \bar{x}(t)| \rightarrow 0$ as $t \rightarrow +\infty$.

Definition 4.14. A solution $\bar{x}(t)$ of (4.6) is uniformly stable if for each $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that if $x(t)$ is a solution of (4.6) and $|x(t_0) - \bar{x}(t_0)| < \delta$ then for some $t_0 \in \mathbb{R}$ then $|x(t) - \bar{x}(t)| < \varepsilon$ for all $t \geq t_0$.

Suppose f is a continuously differentiable function such that

$$\dot{x} = f(x) \quad (4.7)$$

generates a continuous dynamical system φ on $\Omega \subseteq \mathbb{R}^n$. Suppose, moreover, that $x_0 \in \Omega$ is an equilibrium point of φ . If x solves (4.7) and we set $u := x - x_0$ and $A := Df(x_0)$, we see that, by the definition of derivative,

$$\dot{u} = f(u + x_0) = f(x_0) + Df(x_0)u + R(u) = Au + R(u), \quad (4.8)$$

where $R(u)/|u| \rightarrow 0$ as $|u| \rightarrow 0$. Because $R(u)$ is small when u is small, it is reasonable to believe that solutions of (4.8) behave similarly to solutions of

$$\dot{u} = Au \quad (4.9)$$

for u near 0. Equivalently, it is reasonable to believe that solutions of (4.7) behave like solutions of

$$\dot{x} = A(x - x_0) \quad (4.10)$$

for x near x_0 . Equation (4.9) (or sometimes (4.10)) is called the linearization of (4.7) at x_0 .

Now, we've defined (several types of) stability for equilibrium solutions of (4.7), but we haven't really given any tools for determining stability. In this section we present one such tool, using the linearized equation(s) discussed above.

Definition 4.15. An equilibrium point x_0 of (4.7) is hyperbolic if none of the eigenvalues of $Df(x_0)$ have zero real part.

If x_0 is hyperbolic, then either all the eigenvalues of $A := Df(x_0)$ have a negative real part or at least one has a positive real part. In the former case, we know that 0 is an asymptotically stable equilibrium solution of (4.9); in the latter case, we know that 0 is an unstable solution of (4.9). The following theorem says that similar things can be said about the nonlinear equation (4.7).

Theorem 4.16 (Principle of Linearized Stability). *If x_0 is a hyperbolic equilibrium solution of (4.7), then x_0 is either unstable or asymptotically stable, and its stability type (with respect to (4.7)) matches the stability type of 0 as an equilibrium solution of (4.9) (where $A := Df(x_0)$).*

This theorem is an immediate consequence of the following two propositions.

Proposition 4.17 (Asymptotic Stability). *If x_0 is an equilibrium point of (4.7) and all the eigenvalues of $A := Df(x_0)$ have a negative real part, then x_0 is asymptotically stable.*

Proposition 4.18 (Instability). *If x_0 is an equilibrium point of (4.7) and some eigenvalue of $A := Df(x_0)$ has a positive real part, then x_0 is unstable.*

A proof of 4.18 is not given here, but a geometric one can be found [4, Proposition (**Instability**) in Chapter III].

Before we prove 4.17, we state and prove a lemma.

Lemma 4.19. *Let V be a finite-dimensional real vector space and let $L \in \mathcal{L}(V, V)$. If all the eigenvalues of L have real part larger than c , then there is an inner product $\langle \cdot, \cdot \rangle$ and an induced norm $\| \cdot \|$ on V such that*

$$\langle v, Lv \rangle \geq c \|v\|^2$$

for every $v \in V$.

Proof. Let $n = \dim V$, and pick $\varepsilon > 0$ so small that all the eigenvalues of L have real parts greater than $c + n\varepsilon$. Choose a basis $\{v_1, \dots, v_n\}$ for V that puts L in modified real canonical form with the off-diagonal 1's replaced by ε , and let $\langle \cdot, \cdot \rangle$ be the inner product associated with this basis (i.e. $\langle v_i, v_j \rangle = \delta_{ij}$), and let $\|\cdot\|$ be the induced norm on V .

Given $v = \sum_{i=1}^n \alpha_i v_i \in V$, note that (if $L = (\ell_{ij})$)

$$\begin{aligned} \langle v, Lv \rangle &= \sum_{i=1}^n \ell_{ii} \alpha_i^2 + \sum_{i=1}^n \sum_{j \neq i} \ell_{ij} \alpha_i \alpha_j \geq \sum_{i=1}^n \ell_{ii} \alpha_i^2 - \sum_{i=1}^n \sum_{j \neq i} \varepsilon \left(\frac{\alpha_i^2 + \alpha_j^2}{2} \right) \\ &\geq \sum_{i=1}^n \ell_{ii} \alpha_i^2 - \sum_{i=1}^n n\varepsilon \alpha_i^2 = \sum_{i=1}^n (\ell_{ii} - n\varepsilon) \alpha_i^2 \geq \sum_{i=1}^n c \alpha_i^2 = c\|v\|^2. \end{aligned}$$

Note that applying this theorem to $-L$ also tells us that, for some inner product,

$$\langle v, Lv \rangle \leq c\|v\|^2 \quad (4.11)$$

if all the eigenvalues of L have real part less than c .

□

Proof. (Asymptotic Stability) Without loss of generality, assume that $x_0 = 0$. Pick $c < 0$ such that all the eigenvalues of A have real parts strictly less than c . Because of the equivalence of norms and because of the lemma, we can work with a norm $\|\cdot\|$ and a corresponding inner product $\langle \cdot, \cdot \rangle$ for which (4.11) holds, with $L = A$. Let $r > 0$ be small enough that $\|R(x)\| \leq -\frac{c}{2\|x\|}$ for all x satisfying $\|x\| \leq r$, and let

$$B_r := \{x \in \Omega \mid \|x\| < r\}.$$

If $x(t)$ is a solution of (4.7) that starts in B_r at time $t = 0$, then as long as $x(t)$ remains in B_r

$$\begin{aligned} \frac{d}{dt} \|x(t)\|^2 &= 2\langle x(t), \dot{x}(t) \rangle = 2\langle x(t), f(x(t)) \rangle \\ &= 2\langle x(t), Ax(t) \rangle + 2\langle x(t), R(x(t)) \rangle \\ &\leq 2c\|x(t)\|^2 + 2\|x(t)\| \cdot \|R(x(t))\| \\ &\leq 2c\|x(t)\|^2 - c\|x(t)\|^2 = c\|x(t)\|^2. \end{aligned}$$

This means that $x(t) \in B_r$ for all $t \geq 0$, and $x(t)$ converges to 0 (exponentially quickly) as $t \rightarrow +\infty$.

□

Another tool for determining the stability of solutions is Lyapunov's direct method. While this method may actually seem rather indirect, it does work directly on the equation in question instead of on its linearization.

We will consider this method for equilibrium solutions of (possibly) nonautonomous equations. Let $\Omega \subseteq \mathbb{R}^n$ be open and contain the origin, and suppose that $f : \mathbb{R} \times \Omega \rightarrow \mathbb{R}^n$ is a continuously differentiable function. Suppose, furthermore, that $f(t, 0) = 0$ for every $t \in \mathbb{R}$, so $x(t) := 0$ is a solution of the equation

$$\dot{x} = f(t, x). \quad (4.12)$$

(The results we obtain in this narrow context can be applied to determine the stability of other constant solutions of (4.12) by translation.) In this section, a subset of Ω that contains the origin in its interior will be called a neighborhood of 0.

Definition 4.20. Suppose that U is neighborhood of 0 and that $W : U \rightarrow \mathbb{R}$ is continuous and satisfies $W(0) = 0$. Then:

- If $W(x) \geq 0$ for every $x \in U$, then W is positive semidefinite.
- If $W(x) > 0$ for every $x \in U \setminus \{0\}$, then W is positive definite.
- If $W(x) \leq 0$ for every $x \in U$, then W is negative semidefinite.
- If $W(x) < 0$ for every $x \in U \setminus \{0\}$, then W is negative definite.

Definition 4.21. Suppose that U is a neighborhood of 0 and that $V : \mathbb{R} \times U \rightarrow \mathbb{R}$ is continuous and satisfies $V(t, 0) = 0$ for every $t \in \mathbb{R}$. Then:

- If there is a positive semidefinite function $W : U \rightarrow \mathbb{R}$ such that $V(t, x) \geq W(x)$ for every $(t, x) \in \mathbb{R} \times U$, then V is positive semidefinite.
- If there is a positive definite function $W : U \rightarrow \mathbb{R}$ such that $V(t, x) \geq W(x)$ for every $(t, x) \in \mathbb{R} \times U$, then V is positive definite.
- If there is a negative semidefinite function $W : U \rightarrow \mathbb{R}$ such that $V(t, x) \leq W(x)$ for every $(t, x) \in \mathbb{R} \times U$, then V is negative semidefinite.
- If there is a negative definite function $W : U \rightarrow \mathbb{R}$ such that $V(t, x) \leq W(x)$ for every $(t, x) \in \mathbb{R} \times U$, then V is negative definite.

Definition 4.22. If $V : \mathbb{R} \times U \rightarrow \mathbb{R}$ is continuously differentiable then its orbital derivative (with respect to (4.12)) is the function $\dot{V} : \mathbb{R} \times U \rightarrow \mathbb{R}$ given by the formula

$$\dot{V}(t, x) := \frac{\partial V}{\partial t}(t, x) + \frac{\partial V}{\partial x}(t, x) \cdot f(t, x).$$

Here $\partial V(t, x)/\partial x$ represents the gradient of the function $V(t, \cdot)$.

Note that if $x(t)$ is a solution of (4.12) then, by the chain rule,

$$\frac{d}{dt}V(t, x(t)) = \dot{V}(t, x(t)).$$

A function whose orbital derivative is always nonpositive is sometimes called a Lyapunov function.

Theorem 4.23 (Lyapunov Stability). *If there is a neighborhood U of 0 and a continuously differentiable positive definite function $V : \mathbb{R} \times U \rightarrow \mathbb{R}$ whose orbital derivative $\dot{V}$ is negative semidefinite, then 0 is a Lyapunov stable solution of (4.12).*

Proof. Let $\varepsilon > 0$ and $t_0 \in \mathbb{R}$ be given. Assume, without loss of generality, that $\overline{B(0, \varepsilon)}$ is contained in U . Pick a positive definite function $W : U \rightarrow \mathbb{R}$ such that $V(t, x) \geq W(x)$ for every $(t, x) \in \mathbb{R} \times U$. Let

$$m := \min\{W(x) \mid |x| = \varepsilon\}.$$

Since W is continuous and positive definite, m is well-defined and positive. Pick $\delta > 0$ small enough that $\delta < \varepsilon$ and

$$\max\{V(t_0, x) \mid |x| \leq \delta\} < m.$$

(Since V is positive definite and continuous, this is possible.)

Now, if $x(t)$ solves (4.12) and $|x(t_0)| < \delta$ then $V(t_0, x(t_0)) < m$, and

$$\frac{d}{dt}V(t, x(t)) = \dot{V}(t, x(t)) \leq 0,$$

for all t , so $V(t, x(t)) < m$ for every $t \geq t_0$. Thus, $W(x(t)) < m$ for every $t \geq t_0$, so, for every $t \geq t_0$, $|x(t)| \neq \varepsilon$. Since $|x(t_0)| < \varepsilon$, this tells us that $|x(t)| < \varepsilon$ for every $t \geq t_0$. □

Theorem 4.24 (Asymptotic Stability). *Suppose that there is a neighborhood U of 0 and a continuously differentiable positive definite function $V : \mathbb{R} \times U \rightarrow \mathbb{R}$ whose orbital derivative $\dot{V}$ is negative definite, and suppose that there is a positive definite function $\bar{W} : U \rightarrow \mathbb{R}$ such that $V(t, x) \leq \bar{W}(x)$ for every $(t, x) \in \mathbb{R} \times U$. Then 0 is an asymptotically stable solution of (4.12).*

Proof. By the previous theorem, 0 is a Lyapunov stable solution of (4.12). Let $t_0 \in \mathbb{R}$ be given. Assume, without loss of generality, that U is compact. By Lyapunov stability, we know that we can choose a neighborhood $\mathcal{U}$ of 0 such that if $x(t)$ is a solution of (4.12) and $x(t_0) \in \mathcal{U}$, then $x(t) \in U$ for every

$t \geq t_0$. We claim that, in fact, if $x(t)$ is a solution of (4.12) and $x(t_0) \in \mathcal{U}$, then $x(t) \rightarrow 0$ as $t \rightarrow +\infty$. Verifying this claim will prove the theorem. Suppose that $V(t, x(t))$ does not converge to 0 as $t \rightarrow +\infty$. The negative definiteness of $\dot{V}$ implies that $V(\cdot, x(\cdot))$ is nonincreasing, so, since $V \geq 0$, there must be a number $c > 0$ such that $V(t, x(t)) \geq c$ for every $t \geq t_0$. Then $\overline{W}(x(t)) \geq c > 0$ for every $t \geq t_0$. Since $\overline{W}(0) = 0$ and $\overline{W}$ is continuous,

$$\inf \{|x(t)| \mid t \geq t_0\} \geq \varepsilon \quad (4.13)$$

for some constant $\varepsilon > 0$. Pick a negative definite function $Y : U \rightarrow \mathbb{R}$ such that $\dot{V}(t, x) \leq Y(x)$ for every $(t, x) \in \mathbb{R} \times U$. The compactness of $U \setminus \overline{B(0, \varepsilon)}$, along with (4.13), implies that

$$\{Y(x(t)) \mid t \geq t_0\}$$

is bounded away from 0. This, in turn, implies that

$$\{\dot{V}(t, x(t)) \mid t \geq t_0\}$$

is bounded away from 0. In other words,

$$\frac{d}{dt}V(t, x(t)) = \dot{V}(t, x(t)) \leq -\delta \quad (4.14)$$

for some constant $\delta > 0$. Clearly, (4.14) contradicts the nonnegativity of V for large t .

That contradiction implies that $V(t, x(t)) \rightarrow 0$ as $t \rightarrow +\infty$. Pick a positive definite function $\tilde{W} : U \rightarrow \mathbb{R}$ such that $V(t, x) \geq \tilde{W}$ for every $(t, x) \in \mathbb{R} \times U$, and note that $\tilde{W}(x(t)) \rightarrow 0$ as $t \rightarrow +\infty$.

Let $r > 0$ be given, and let

$$w_r = \min \{\tilde{W}(p) \mid p \in U \setminus \overline{B(0, r)}\}, \quad (4.15)$$

which is defined and positive by the compactness of U and the continuity and positive definiteness of $\tilde{W}$. Since $\tilde{W}(x(t)) \rightarrow 0$ as $t \rightarrow +\infty$, there exists T such that $\tilde{W}(x(t)) < w_r$ for every $t > T$. Thus, for $t > T$, it must be the case that $x(t) \in \overline{B(0, r)}$. Hence, 0 is asymptotically stable. $\square$

It may seem strange that we need to bound V by a time-independent, positive definite function W from above. Indeed, some textbooks contain asymptotic stability theorems omitting this hypothesis. There is a counterexample by Massera that demonstrates the necessity of the hypothesis, proven in [4, Exercise 14 in Chapter III].

We have talked about two different tools that can be used to prove that an equilibrium point x_0 of an autonomous system $\dot{x} = f(x)$ is asymptotically stable: linearization and Lyapunov's direct method. One might ask which of these methods is better. Certainly, linearization seems easier to apply because of its straightforward nature: Compute the eigenvalues of $Df(x_0)$. The direct method requires finding an appropriate Lyapunov function, which doesn't seem so straightforward. But, in fact, anytime linearization works, a simple Lyapunov function works, as well.

Definition 4.25. A matrix $A \in \mathcal{M}(n, \mathbb{C})$ is called a normal matrix if

$$AA^* = A^*A.$$

Remark 4.26. The set of normal matrices includes all the Hermitian matrices ($A^* = A$), the skew-Hermitian matrices ($A^* = -A$), and the unitary matrices ($AA^* = A^*A = I$).

Here is an alternate characterization of normal matrices.

Theorem 4.27. A matrix $A \in \mathcal{M}(n, \mathbb{C})$ is a normal matrix if and only if

$$\|Ax\|_2 = \|A^*x\|_2 \quad \text{for all } x \in \mathbb{C}^n.$$

In short, the Spectral Theorem for normal matrices basically states that a matrix A is normal if and only if it is unitarily diagonalizable — i.e., there exists a unitary matrix U and a diagonal matrix D such that $A = UDU^*$. It is important to note that the latter is equivalent to saying that there exists an orthonormal basis (the columns of U) of eigenvectors of A (the corresponding eigenvalues being the diagonal elements of D). Additionally, the following result provides an easy-to-check necessary and sufficient condition for normality.

Theorem 4.28 (Spectral Theorem). Given $A \in \mathcal{M}(n, \mathbb{C})$, the following statements are equivalent:

1. A is normal.
2. A is unitarily diagonalizable.
3. $\sum_{1 \leq i, j \leq n} |a_{i,j}|^2 = \sum_{1 \leq i \leq n} |\lambda_i|^2$, where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A , counting multiplicities.

Corollary 4.29. If a matrix $A \in \mathcal{M}(n, \mathbb{C})$ is Hermitian, then A is unitarily diagonalizable and its eigenvalues are real.

Lemma 4.30. Let $f : (E_1, d_1) \rightarrow (E_2, d_2)$ a function. f is a contractive mapping if and only if there is a $k \in [0, 1)$ such that for every x, y $d_2(f(x), f(y)) \leq kd_1(x, y)$; f is a contraction mapping when $E_1 = E_2$ and $d_1 = d_2$.

Remark 4.31. If $E_1 = E_2$ and $d(x, y) = \|x - y\|_2$, then $\forall x, y \ \|A(x - y)\|_2 \leq k\|x - y\|_2$; the latter is equivalent to saying

$$\sup_{\|u\|=1} \|Au\|_2 \leq k,$$

that is $\exists k < 1$ such that $\|A\|_2 \leq k$ and $\|A\|_2 = \rho(AA^*)$, where $\rho(B)$ is the spectral radius of B .

Theorem 4.32 (Banach fixed-point theorem). *Let (X, d) be a non-empty complete metric space with a contraction mapping $T : X \rightarrow X$. Then T admits a unique fixed-point $\tilde{x} \in X$, i.e. $T(\tilde{x}) = \tilde{x}$. Furthermore, $\tilde{x}$ can be found as follows: start with an arbitrary element $x_0 \in X$ and define a sequence $(x_n)_{n \in \mathbb{N}}$ by $x_n = T(x_{n-1})$ for $n \geq 1$. Then $\lim_{n \rightarrow +\infty} x_n = \tilde{x}$.*

Proof. A proof of this fact can be found in [4, Section 1.3]. □

Definition 4.33. A strongly continuous semigroup on a Banach space X is a map $T : \mathbb{R}^+ \rightarrow \mathcal{L}(X)$ (the space of bounded operators on X) such that:

1. $T(0) = I$ (identity operator on X).
2. $\forall t, s \geq 0: T(t + s) = T(t)T(s)$.
3. $\forall x_0 \in X: \|T(t)x_0 - x_0\| \rightarrow 0$, as $t \rightarrow 0^+$.

The first two axioms are algebraic, and state that T is a representation of the semigroup $(\mathbb{R}^+, +)$; the last is topological, and states that the map T is continuous in the strong operator topology.

Definition 4.34. A uniformly continuous semigroup is a strongly continuous semigroup T such that $\lim_{t \rightarrow 0^+} \|T(t) - I\| = 0$ holds. In this case, the infinitesimal generator A of T is bounded and we have $\mathcal{D}(A) = X$. Moreover,

$$T(t) = e^{At} = \sum_k \frac{A^k}{k!} t^k.$$

Remark 4.35. $\mathcal{D}(A)$ is the set of $x \in X$ for which this limit does exist; $\mathcal{D}(A)$ is a linear subspace and A is linear on his domain. The domain is dense in X .

Appendix B

5.1 Time-dependent reproduction number

The R_t index, or the time-dependent reproduction number, is a key parameter in describing the spread of an infectious disease. It represents the average number of secondary infections caused by a single infected individual at a given time t . This value evolves over time based on the disease dynamics and any implemented control measures.

The R_t index in such a model is typically derived from the basic reproduction number R_0 , adjusted for the time-dependent dynamics of the susceptible population. It can be expressed as:

$$R_t = R_0 \frac{S(t)}{N},$$

where R_0 is the basic reproduction number (when the entire population is susceptible), that is in our case

$$R_0 = \lambda \left[\frac{\beta}{\mu_{IR} + \mu_{ID}} + \frac{1 - \beta}{\mu_{CR} + \mu_{CD}} \right]$$

This formula captures the combined contributions of both symptomatic and asymptomatic individuals to the basic reproduction number. The first term $\frac{\beta}{\mu_{IR} + \mu_{ID}}$ represents the contribution of symptomatic individuals (I) to the spread of the disease. The second term $\frac{1 - \beta}{\mu_{CR} + \mu_{CD}}$ represents the contribution of carriers (C), who are asymptomatic but still infectious. Finally, λ (transmission rate) scales the total contribution.

Although in our code we used another approach: the “Rolling Window Method”. It estimates R_t using a rolling window of observed new infections. It calculates the ratio of new cases in successive time intervals within a fixed “serial interval”, representing the average time between successive generations of infections.

We choose this method since it relies directly on the observed data for new infections and reflects the short-term reproduction number based on observed trends, while the first formula depends on accurate parameter estimation and model assumptions.

5.1.1 Interpretation of reproduction number

- If $R_t > 1$ the epidemic is expanding (each infected individual infects more than one person).
- If $R_t = 1$ the epidemic is stable (the number of new cases remains constant).
- If $R_t < 1$ the epidemic is declining (each infected individual infects fewer than one person).

Using real COVID-19 data in Italy we tried to reproduce the R_t index through a graph:

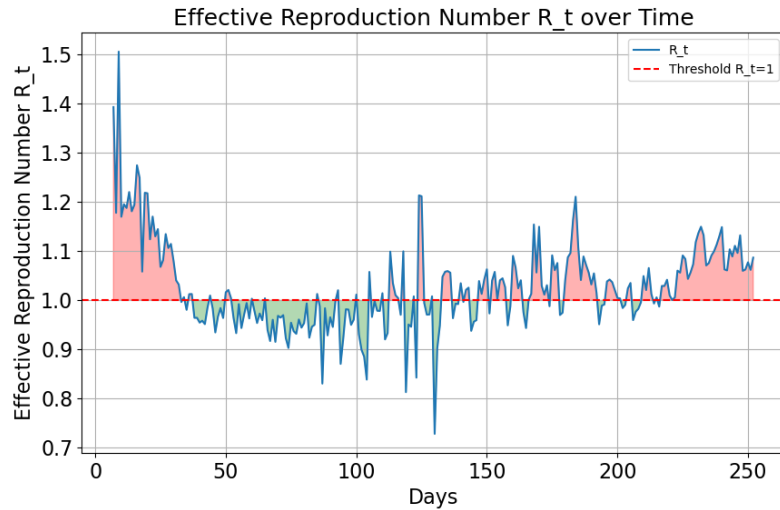


Figure 5.1: R_t Index Italy

We have represented the critical threshold with a dotted line and above it, the red color indicates the situation of expansion of the epidemic, while below it

the green color indicates its decline. This graph should faithfully reproduce the epidemic situation in Italy from February to the end of October 2020. The following is the code used:

```
1 # Define the serial interval (time between successive cases)
2 serial_interval = 7 # Example value, adjust as needed
```

The serial interval in the epidemiology of communicable (infectious) diseases is the time between successive cases in a chain of transmission. It could in principle be estimated by the time interval between infection and subsequent transmission. This defines the size of the "rolling window" used to calculate R_t .

```
1 # Estimate R_t using a rolling window method
2 R_t_italy = []
3 for t in range(len(new_infected_real) - serial_interval):
4     current_window = new_infected_real[t:t + serial_interval]
5     if np.sum(current_window) > 0:
6         R_t_italy.append(np.sum(current_window[1:]) /
7             ↪ np.sum(current_window[:-1]))
8     else:
9         R_t_italy.append(np.nan)
```

`R_t_italy`: An empty list to store the calculated (R_t) values.

`for t in range(...)`: Loop through the time series of new infections.

`current_window`: A slice of the time series of length equal to the serial interval.

`np.sum(current_window) > 0`: Check if the sum of the current window is greater than zero to avoid division by zero.

`R_t_italy.append(...)`: Calculate (R_t) as the ratio of the sum of new infections in the latter part of the window to the sum in the earlier part.

```
1 # Pad the R_t list to match the length of the time series
2 R_t = [np.nan] * serial_interval + R_t_italy
3
4 # Convert R_t to a NumPy array
5 R_t = np.array(R_t)
```

Here we pad the beginning of the (R_t) list with NaNs to match the length of the original time series and then we convert the list to a NumPy array for easier manipulation and plotting.

```
1 # Plot the results for R_t over time
2 plt.figure(figsize=(10, 6))
```

```

3 plt.plot(R_t, label='R_t')
4 plt.axhline(y=1, color='r', linestyle='--', label='Threshold
  ↳ R_t=1')
5 plt.fill_between(t_data, R_t, 1, where=(R_t > 1),
  ↳ interpolate=True, color='red', alpha=0.3)
6 plt.fill_between(t_data, R_t, 1, where=(R_t <= 1),
  ↳ interpolate=True, color='green', alpha=0.3)
7 plt.xlabel('Days', fontsize=16) # Adjust the font size for the
  ↳ x-axis label
8 plt.ylabel('Effective Reproduction Number R_t', fontsize=16) #
  ↳ Adjust the font size for the y-axis label
9 plt.title('Effective Reproduction Number R_t over Time',
  ↳ fontsize=18) # Adjust the font size for the title
10 plt.xticks(fontsize=16) # Adjust the font size for the x-axis
  ↳ tick labels
11 plt.yticks(fontsize=16) # Adjust the font size for the y-axis
  ↳ tick labels
12 plt.legend()
13 plt.grid(True)
14 plt.show()

```

This code calculates and visualizes the effective reproduction number (R_t) over time, helping to understand the dynamics of an infectious disease.

Bibliography

- [1] Fred Brauer. Compartmental models in epidemiology. In *Mathematical epidemiology*, volume 1945 of *Lecture Notes in Math.*, pages 19–79. Springer, Berlin, 2008.
- [2] Fred Brauer and P. van den Driessche. Some directions for mathematical epidemiology. In *Dynamical systems and their applications in biology (Cape Breton Island, NS, 2001)*, volume 36 of *Fields Inst. Commun.*, pages 95–112. Amer. Math. Soc., Providence, RI, 2003.
- [3] S. Eubank, I. Eckstrand, B. Lewis, S. Venkatramanan, M. Marathe, and C. L. Barrett. Commentary on Ferguson, et al., “Impact of non-pharmaceutical interventions (NPIs) to reduce COVID-19 mortality and healthcare demand”. *Bull. Math. Biol.*, 82(4):Paper No. 52, 7, 2020.
- [4] C.P. Grant. *Theory of Ordinary Differential Equations*. CreateSpace Independent Publishing Platform, 2014.
- [5] P. J. Harris and B. E. J. Bodmann. A mathematical model for simulating the spread of a disease through a country divided into geographical regions with different population densities. *J. Math. Biol.*, 85(4):Paper No. 32, 20, 2022.
- [6] Matt J. Keeling and Pejman Rohani. *Modeling infectious diseases in humans and animals*. Princeton University Press, Princeton, NJ, 2008.
- [7] Francesca Maggioni, Daniel Faccini, Federico Gheza, Filippo Manelli, and Graziella Bonetti. Machine learning based classification models for COVID-19 patients. In *Operations research for health care in red zone*, volume 10 of *AIRO Springer Ser.*, pages 35–46. Springer, Cham, [2023] ©2023.
- [8] J. D. Murray. *Mathematical biology. II*, volume 18 of *Interdisciplinary Applied Mathematics*. Springer-Verlag, New York, third edition, 2003. Spatial models and biomedical applications.

- [9] Vasileios E. Papageorgiou and George Tsaklidis. A stochastic particle extended SEIRS model with repeated vaccination: application to real data of COVID-19 in Italy. *Math. Methods Appl. Sci.*, 47(7):6504–6538, 2024.
- [10] Frederico José Ribeiro Pelogia, Henrique Mohallem Paiva, and Roberson Saraiva Polli. Multi-wave modelling and short-term prediction of ICU bed occupancy by patients with COVID-19 in regions of Italy. *Math. Model. Nat. Phenom.*, 19:Paper No. 13, 17, 2024.
- [11] Benjamin Seibold. *M-Matrices in Meshless Finite Difference Methods*. PhD thesis, University of Kaiserslautern, Kaiserslautern, Germany, 2006.